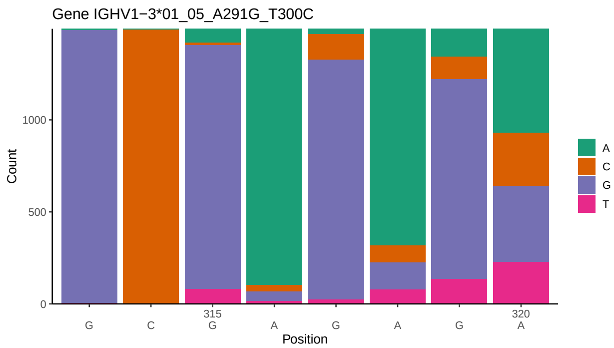
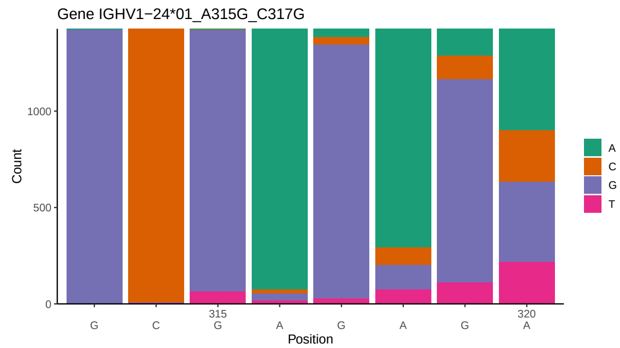
OGRDBstats Report

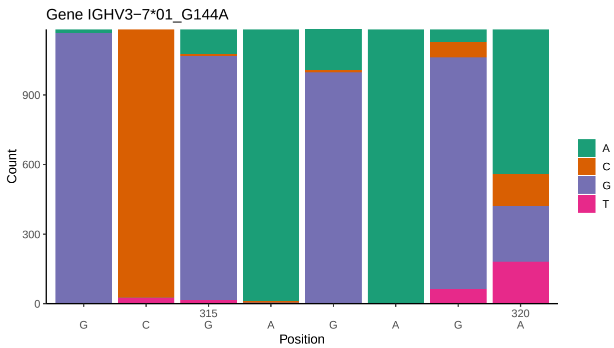
Contents

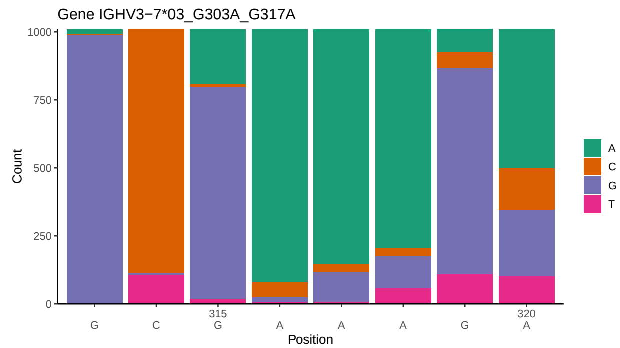
1 Novel sequence analysis

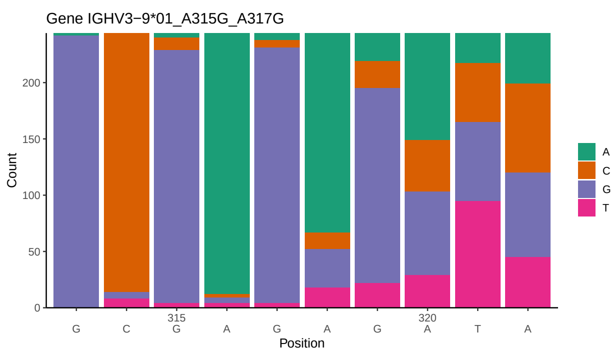
1.1 End-nucleotide composition

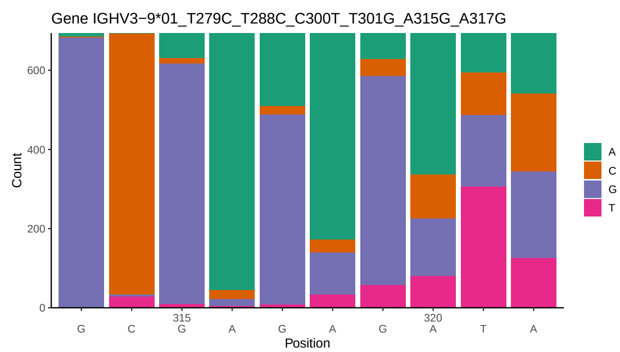


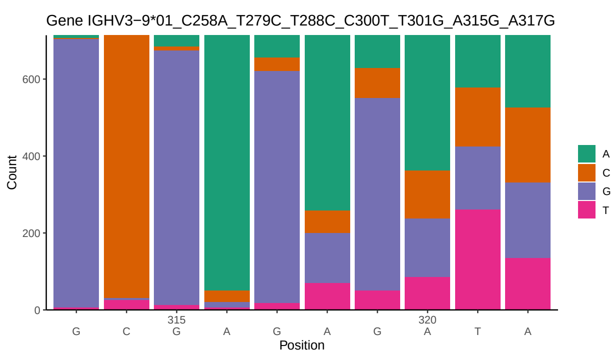


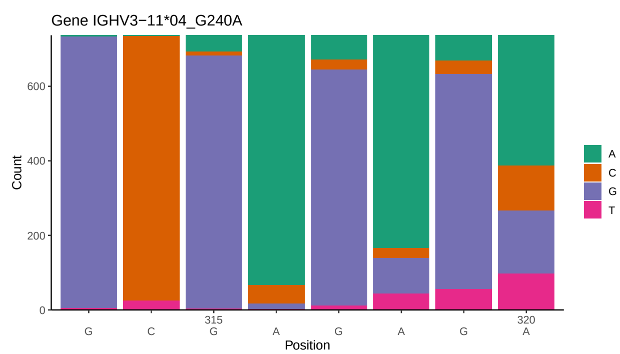


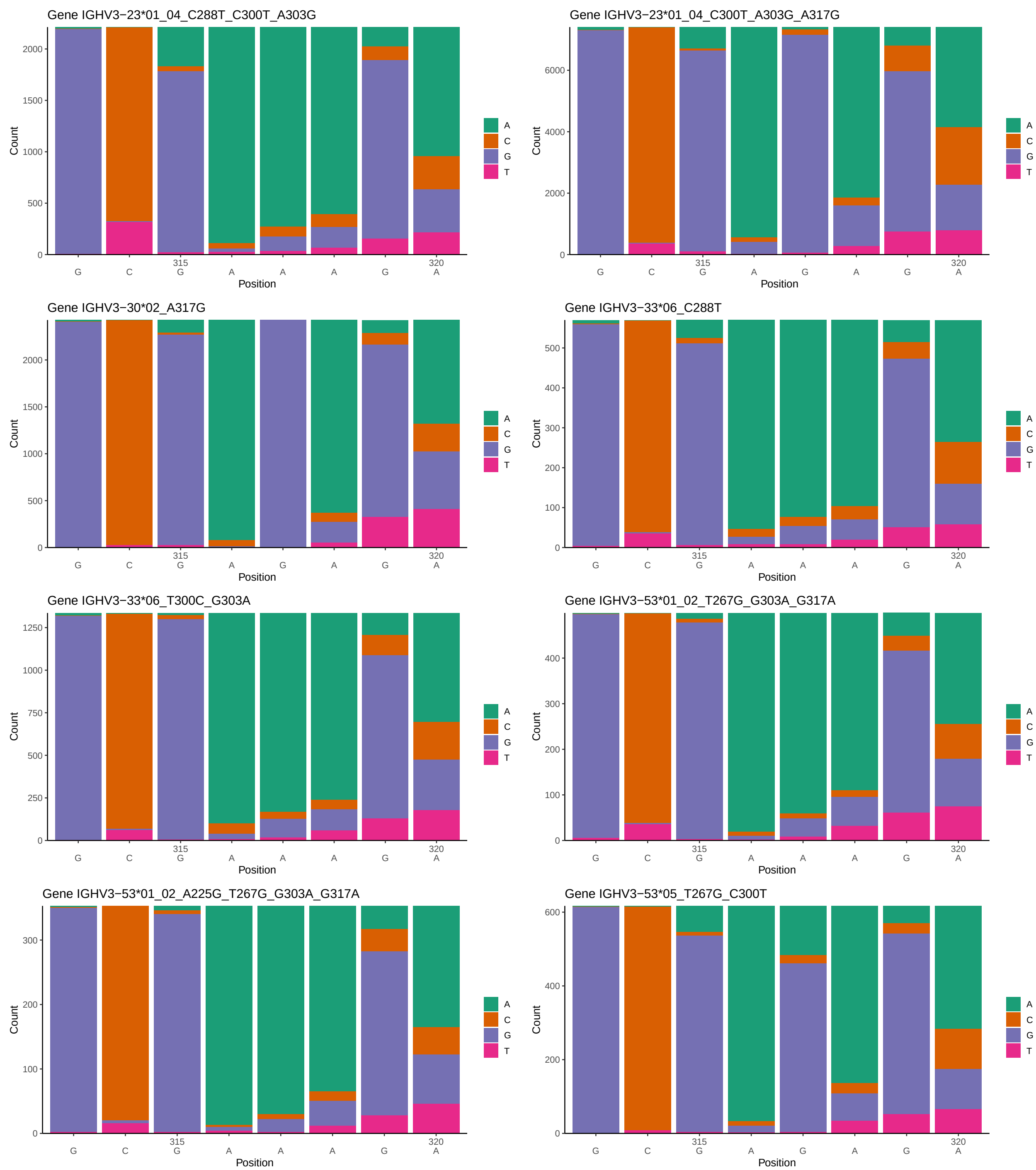








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Count
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500
1000
1500
2000
Position
G C 315 G A A A G 320 A
A C G T
Gene IGHV3−23*01_04_C300T_A303G_A317G
Count
0
2000
4000
6000
Position
G C 315 G A G A G 320 A
A C G T
Gene IGHV3−30*02_A317G
Count
0
500
1000
1500
2000
Position
G C 315 G A G A G 320 A
A C G T
Gene IGHV3−33*06_C288T
Count
0
100
200
300
400
500
Position
G C 315 G A A A G 320 A
A C G T
Gene IGHV3−33*06_T300C_G303A
Count
0
250
500
750
1000
1250
Position
G C 315 G A A A G 320 A
A C G T
Gene IGHV3−53*01_02_T267G_G303A_G317A
Count
0
100
200
300
400
Position
G C 315 G A A A G 320 A
A C G T
Gene IGHV3−53*01_02_A225G_T267G_G303A_G317A
Count
0
100
200
300
Position
G C 315 G A A A G 320 A
A C G T
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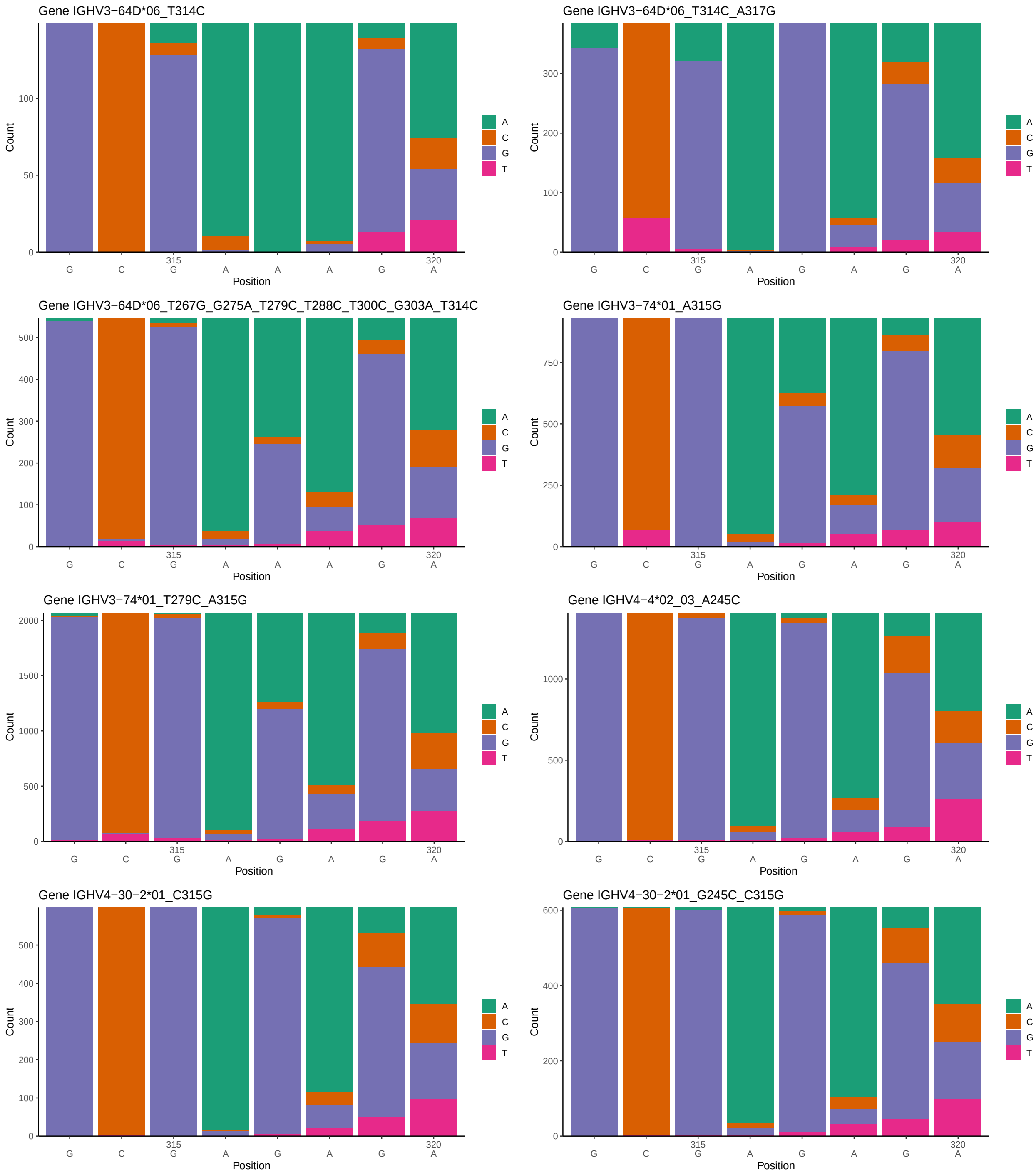
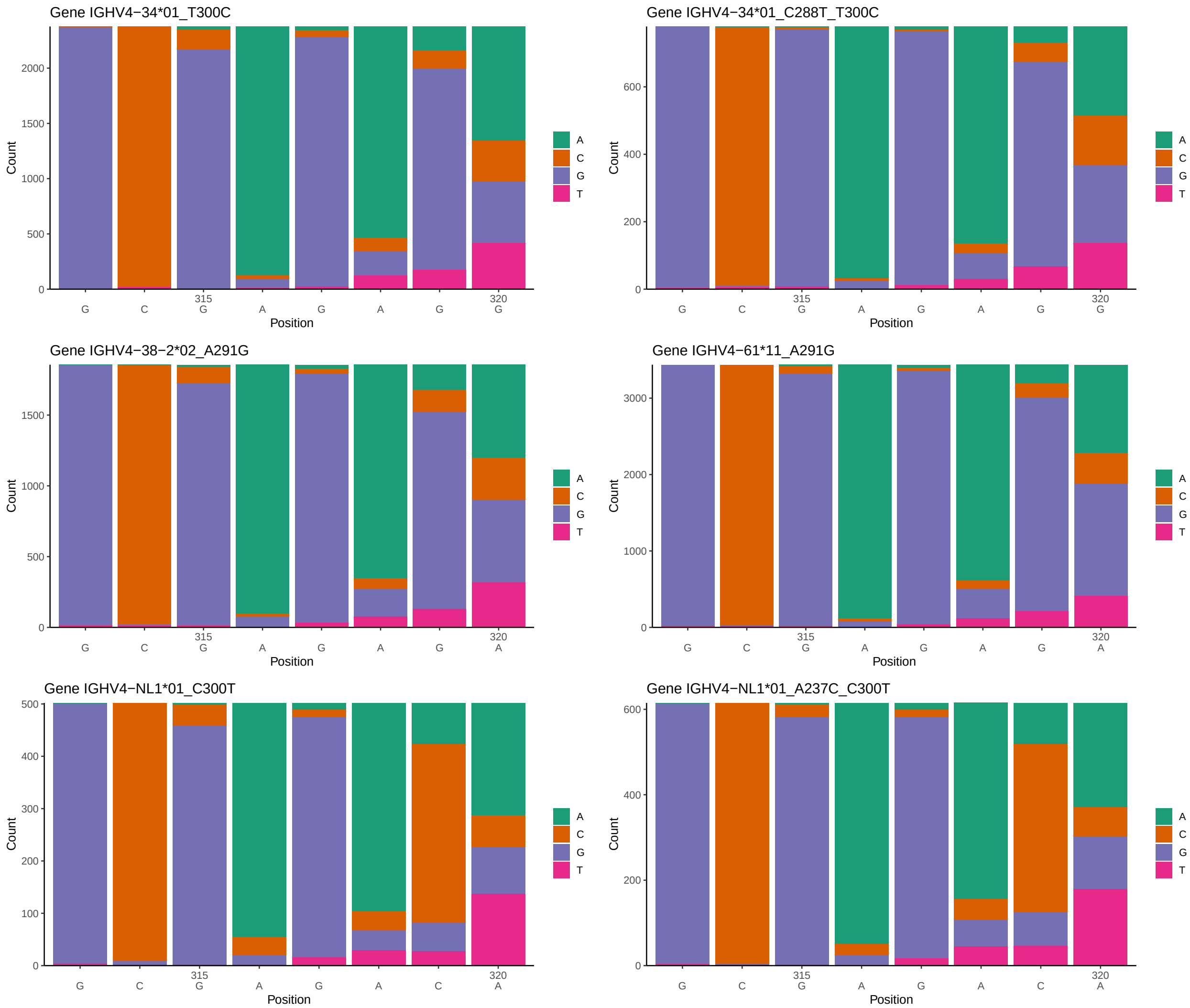
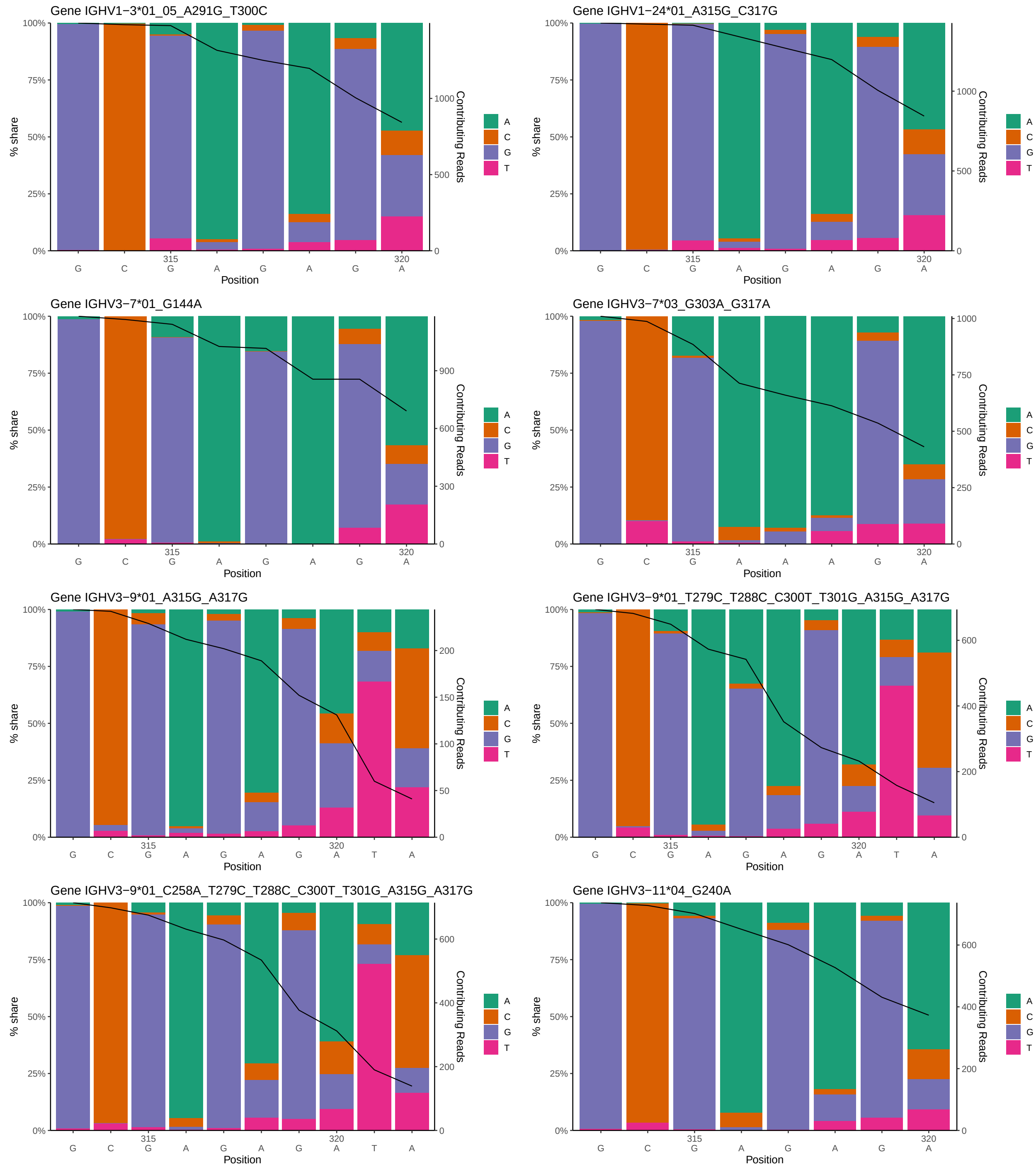
Count
0
200
400
600
Position
G C 315 G A G A G 320 A
A C G T

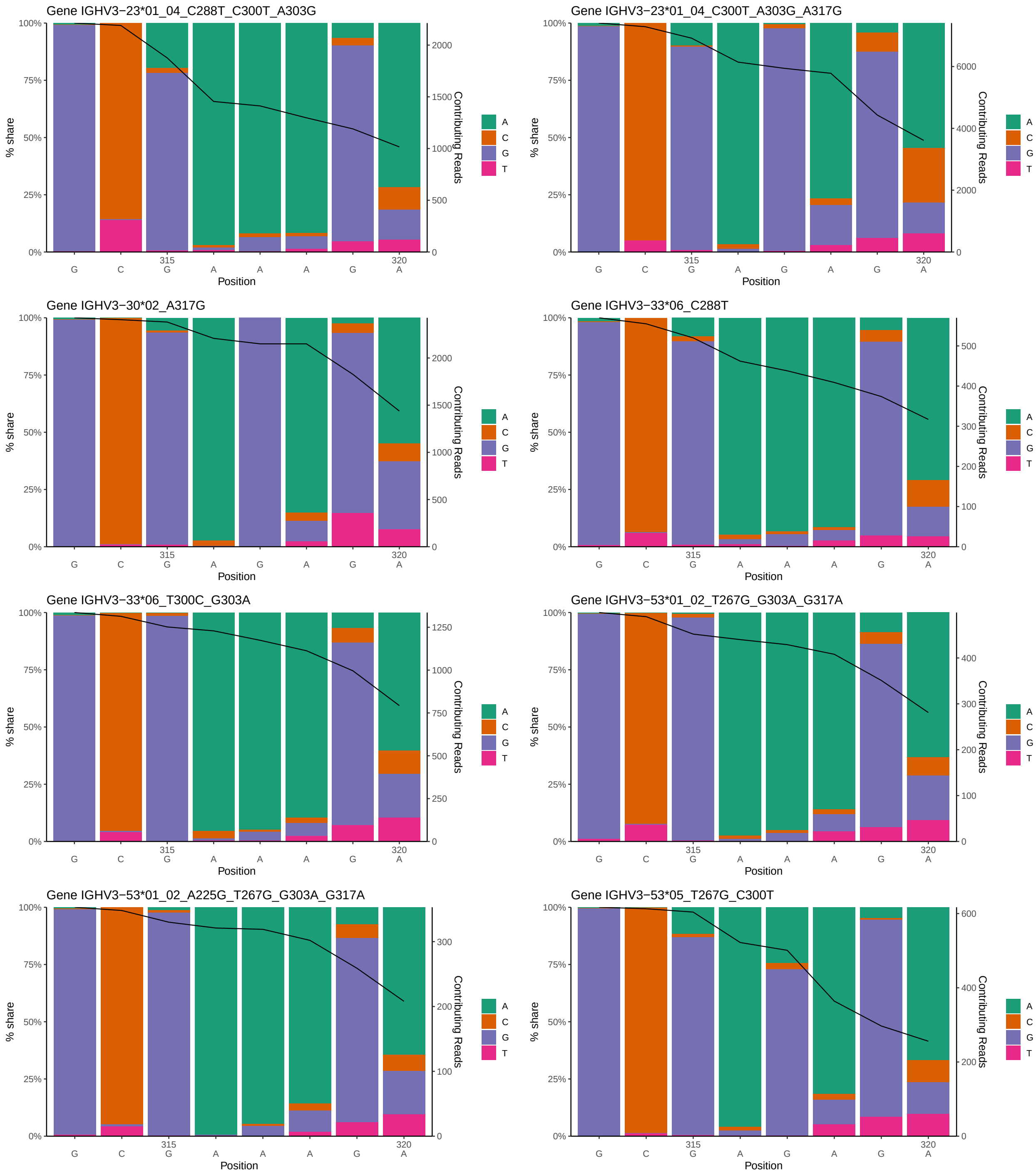


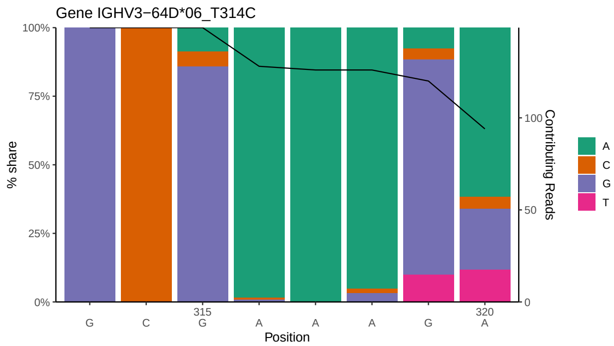
Gene IGHV3−64D*06_T314C
Gene IGHV3−64D*06_T314C_A317G
Gene IGHV3−64D*06_T267G_G275A_T279C_T288C_T300C_G303A_T314C
Gene IGHV3−74*01_A315G
Gene IGHV3−74*01_T279C_A315G
Gene IGHV4−4*02_03_A245C
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−2*01_G245C_C315G
Count
Position
A
C
G
T

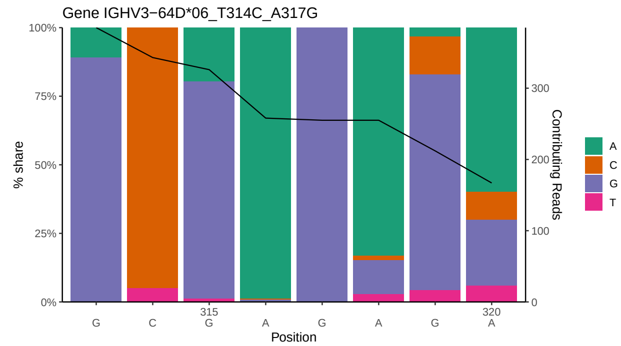

Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_C288T_T300C
Gene IGHV4−38−2*02_A291G
Gene IGHV4−61*11_A291G
Gene IGHV4−NL1*01_C300T
Gene IGHV4−NL1*01_A237C_C300T
Count
Position
315
320
G C G A G A G G
G C G A G A G A
G C G A G A C A
A
C
G
T

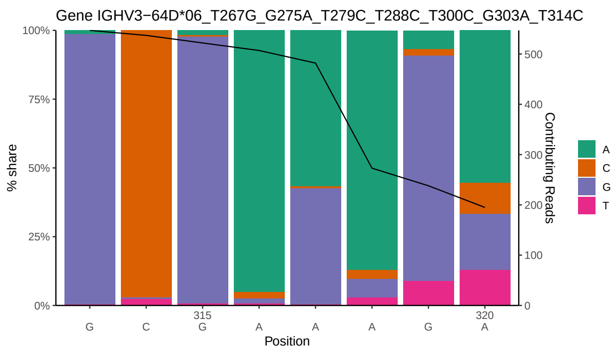
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

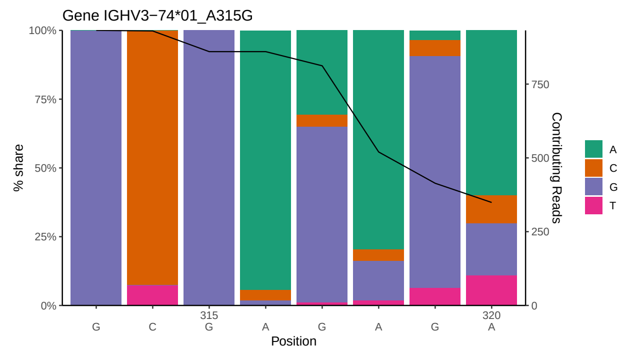


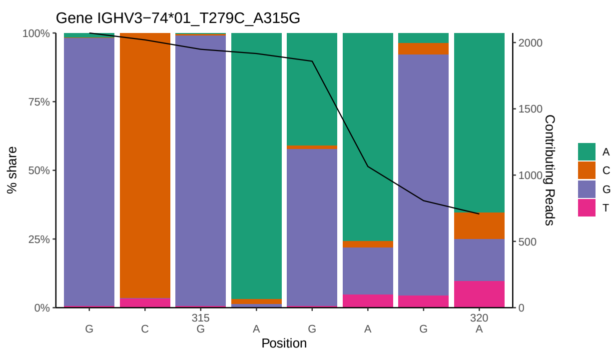

Gene IGHV3−23*01_04_C288T_C300T_A303G
Gene IGHV3−23*01_04_C300T_A303G_A317G
Gene IGHV3−30*02_A317G
Gene IGHV3−33*06_C288T
Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−53*01_02_T267G_G303A_G317A
Gene IGHV3−53*01_02_A225G_T267G_G303A_G317A
Gene IGHV3−53*05_T267G_C300T
% share
100%
75%
50%
25%
0%
Contributing Reads
Position
315
320
G C G A A A G A
G C G A G A G A
A
C
G
T

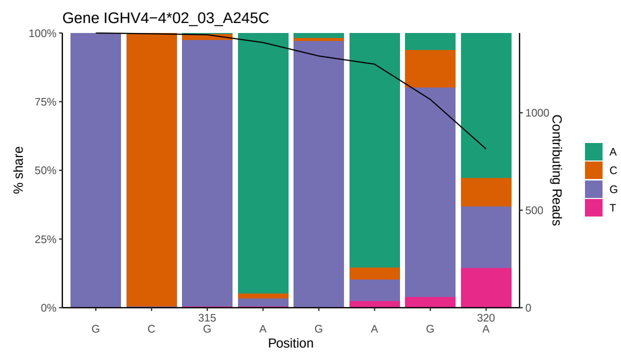

Gene IGHV3−64D*06_T314C
% share
100%
75%
50%
25%
0%
Contributing Reads
100
50
0
G C G A A A G A
315
320
Position
A
C
G
T

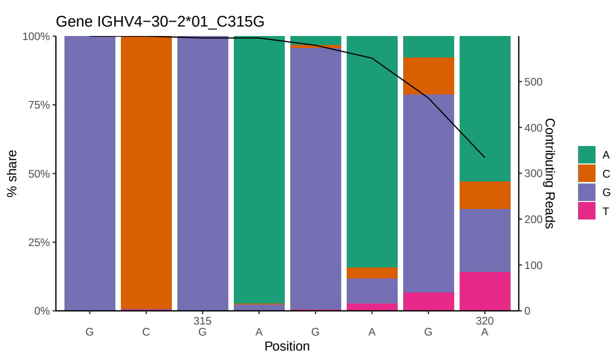

Gene IGHV3−64D*06_T314C_A317G
% share
100%
75%
50%
25%
0%
Contributing Reads
300
200
100
0
G C G A G A G A
315
320
Position
A
C
G
T

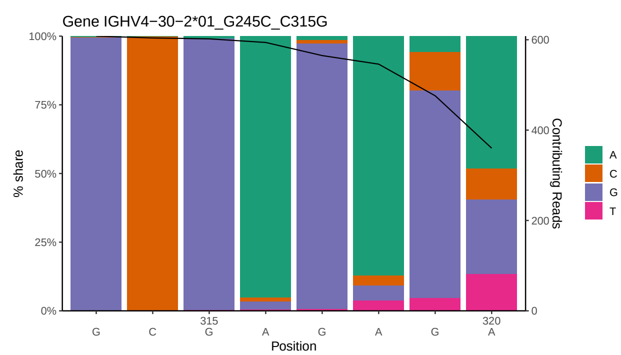

Gene IGHV3−64D*06_T267G_G275A_T279C_T288C_T300C_G303A_T314C
% share
100%
75%
50%
25%
0%
Contributing Reads
500
400
300
200
100
0
G C G A A A G A
315
320
Position
A
C
G
T

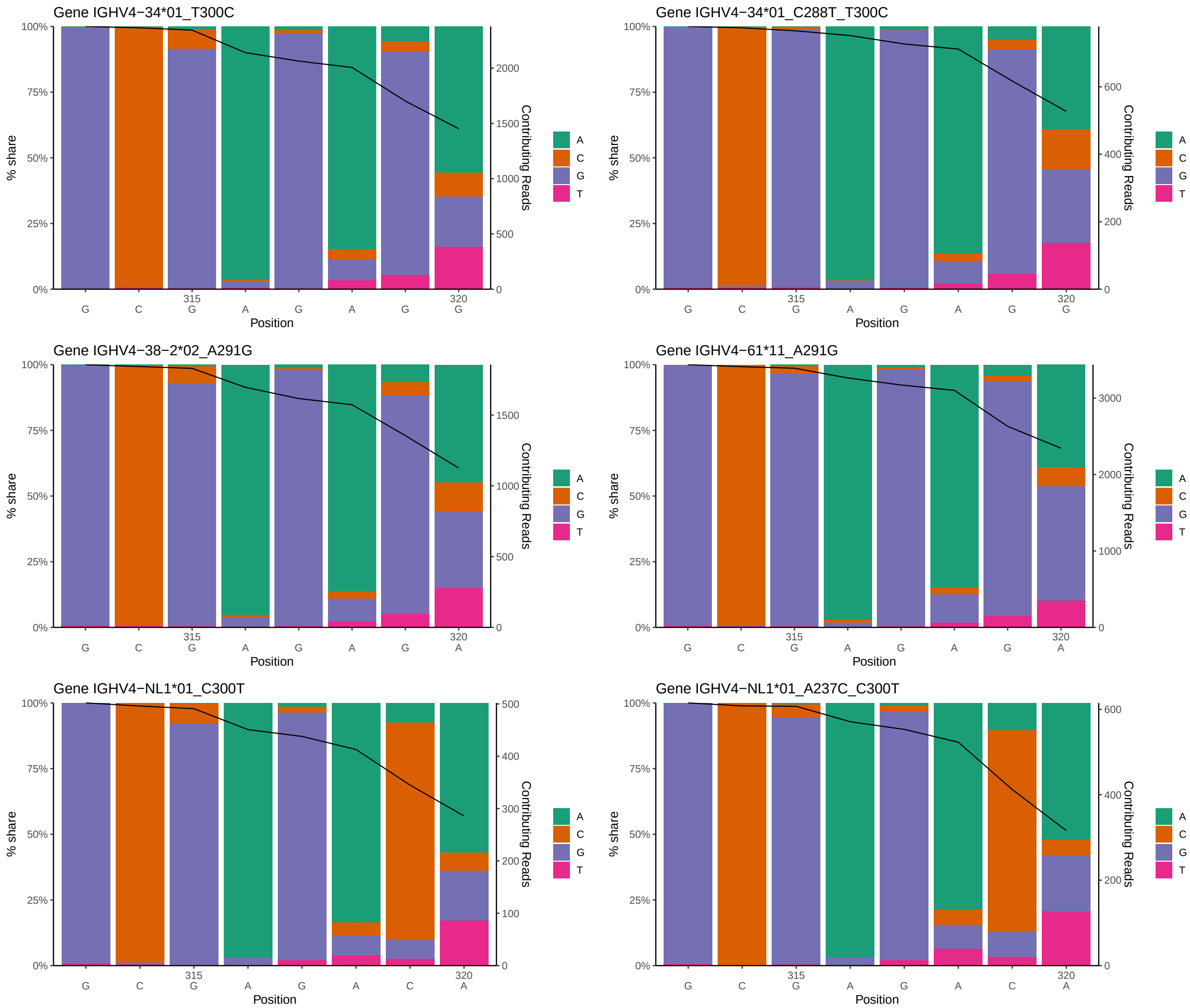

Gene IGHV3−74*01_A315G
% share
100%
75%
50%
25%
0%
Contributing Reads
750
500
250
0
G C G A G A G A
315
320
Position
A
C
G
T


Gene IGHV3−74*01_T279C_A315G
% share
100%
75%
50%
25%
0%
Contributing Reads
2000
1500
1000
500
0
G C G A G A G A
315
320
Position
A
C
G
T

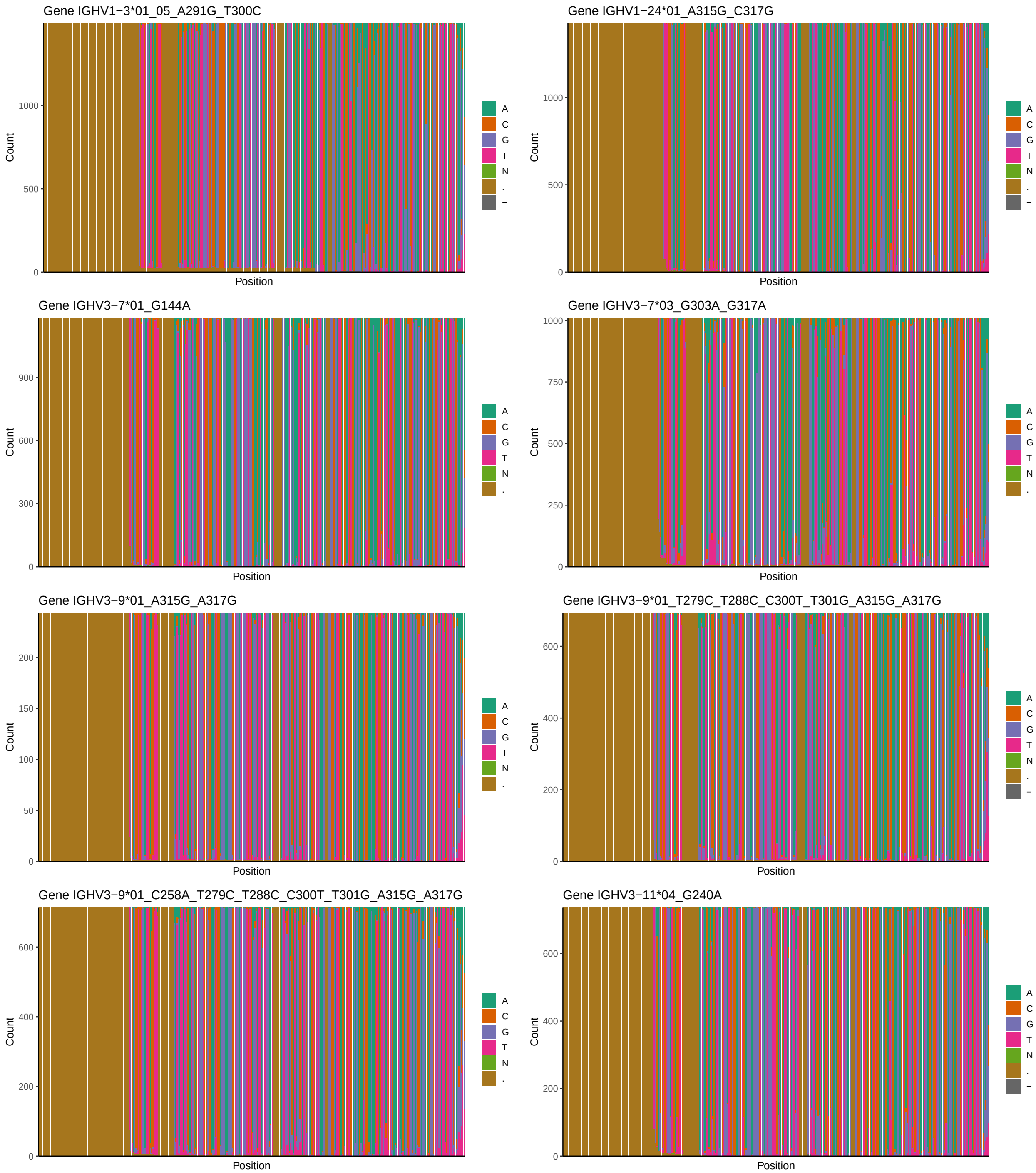

Gene IGHV4−4*02_03_A245C
% share
100%
75%
50%
25%
0%
Contributing Reads
1000
500
0
G C G A G A G A
315
320
Position
A
C
G
T

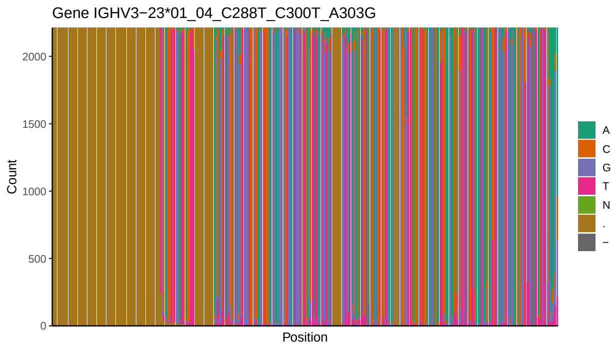

Gene IGHV4−30−2*01_C315G
% share
100%
75%
50%
25%
0%
Contributing Reads
500
400
300
200
100
0
G C G A G A G A
315
320
Position
A
C
G
T

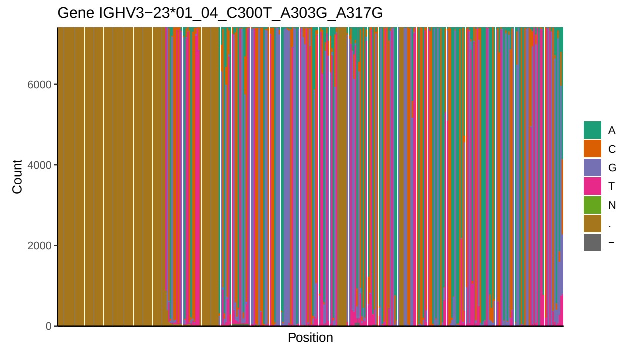

Gene IGHV4−30−2*01_G245C_C315G
% share
100%
75%
50%
25%
0%
Contributing Reads
600
400
200
0
G C G A G A G A
315
320
Position
A
C
G
T

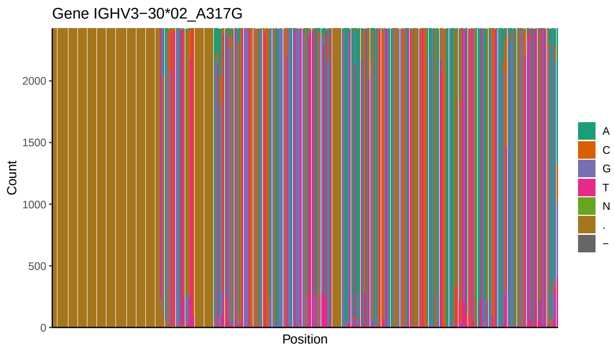

Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_C288T_T300C
Gene IGHV4−38−2*02_A291G
Gene IGHV4−61*11_A291G
Gene IGHV4−NL1*01_C300T
Gene IGHV4−NL1*01_A237C_C300T
% share
Contributing Reads
Position
A
C
G
T

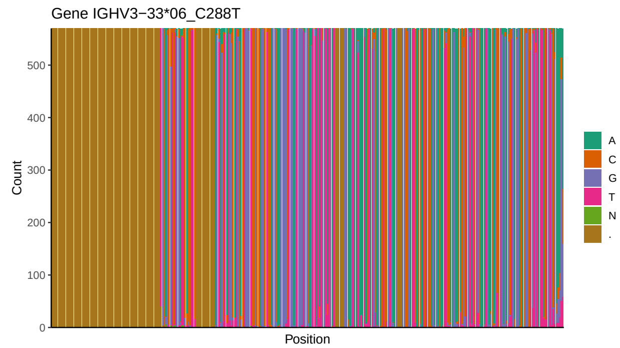
1.3 Whole-sequence composition of each assigned read

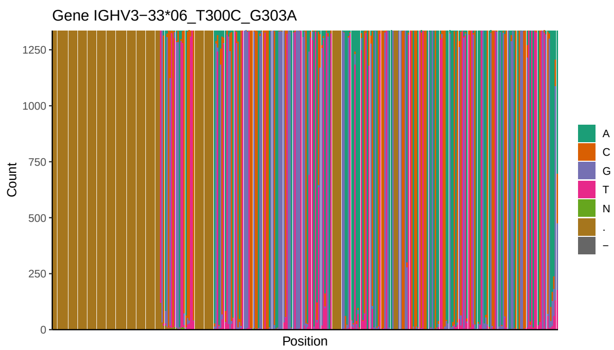



Gene IGHV3−23*01_04_C288T_C300T_A303G
Count
0
500
1000
1500
2000
Position
A
C
G
T
N
.
−


Gene IGHV3−23*01_04_C300T_A303G_A317G
Count
0
2000
4000
6000
Position
A
C
G
T
N
.
−

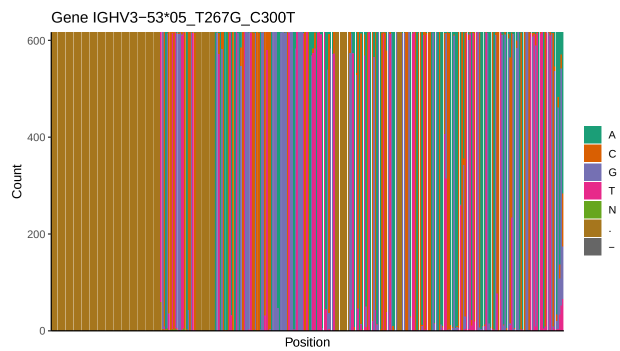

Gene IGHV3−30*02_A317G
Count
0
500
1000
1500
2000
Position
A
C
G
T
N
.
−

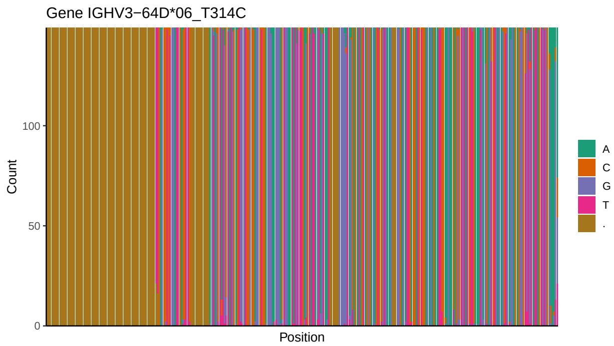
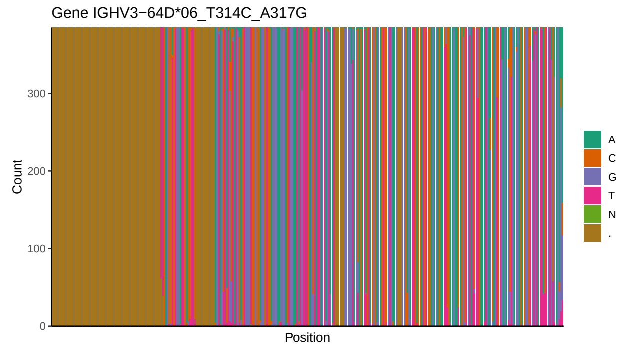

Gene IGHV3−33*06_C288T
Count
0
100
200
300
400
500
Position
A
C
G
T
N
.


Gene IGHV3−33*06_T300C_G303A
Count
0
250
500
750
1000
1250
Position
A
C
G
T
N
.
−


Gene IGHV3−53*01_02_T267G_G303A_G317A
Count
0
100
200
300
400
Position
A
C
G
T
N
.


Gene IGHV3−53*01_02_A225G_T267G_G303A_G317A
Count
0
100
200
300
Position
A
C
G
T
N
.


Gene IGHV3−53*05_T267G_C300T
Count
0
200
400
600
Position
A
C
G
T
N
.
−


Gene IGHV3−64D*06_T314C
Count
100
50
0
Position
A
C
G
T
.

Gene IGHV3−64D*06_T314C_A317G
Count
300
200
100
0
Position
A
C
G
T
N
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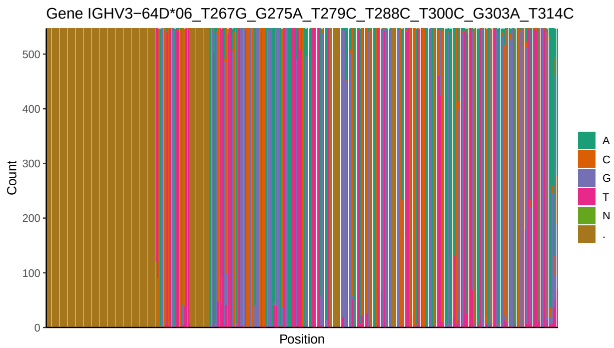
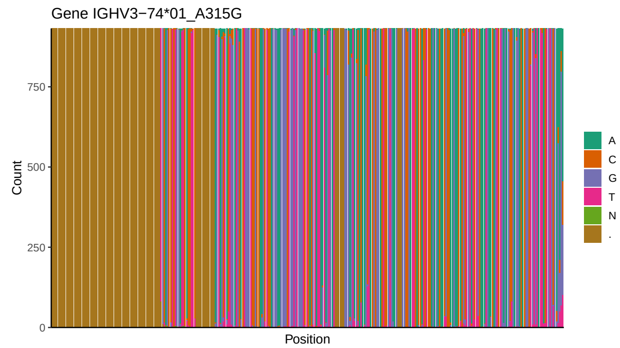
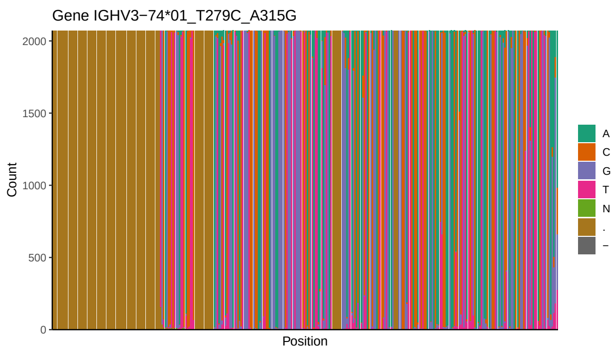
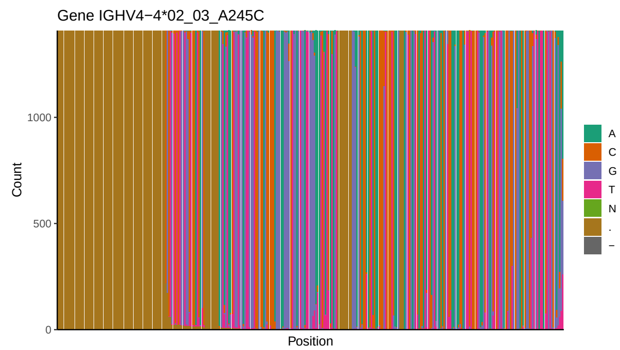
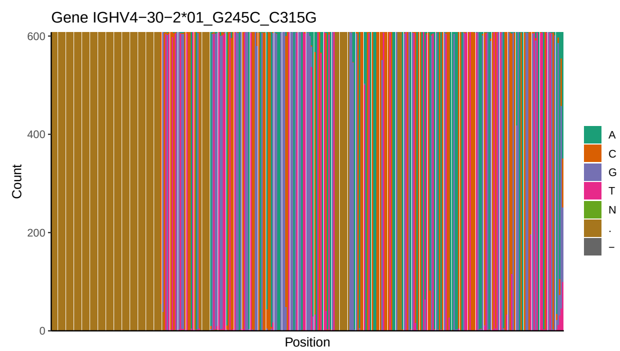
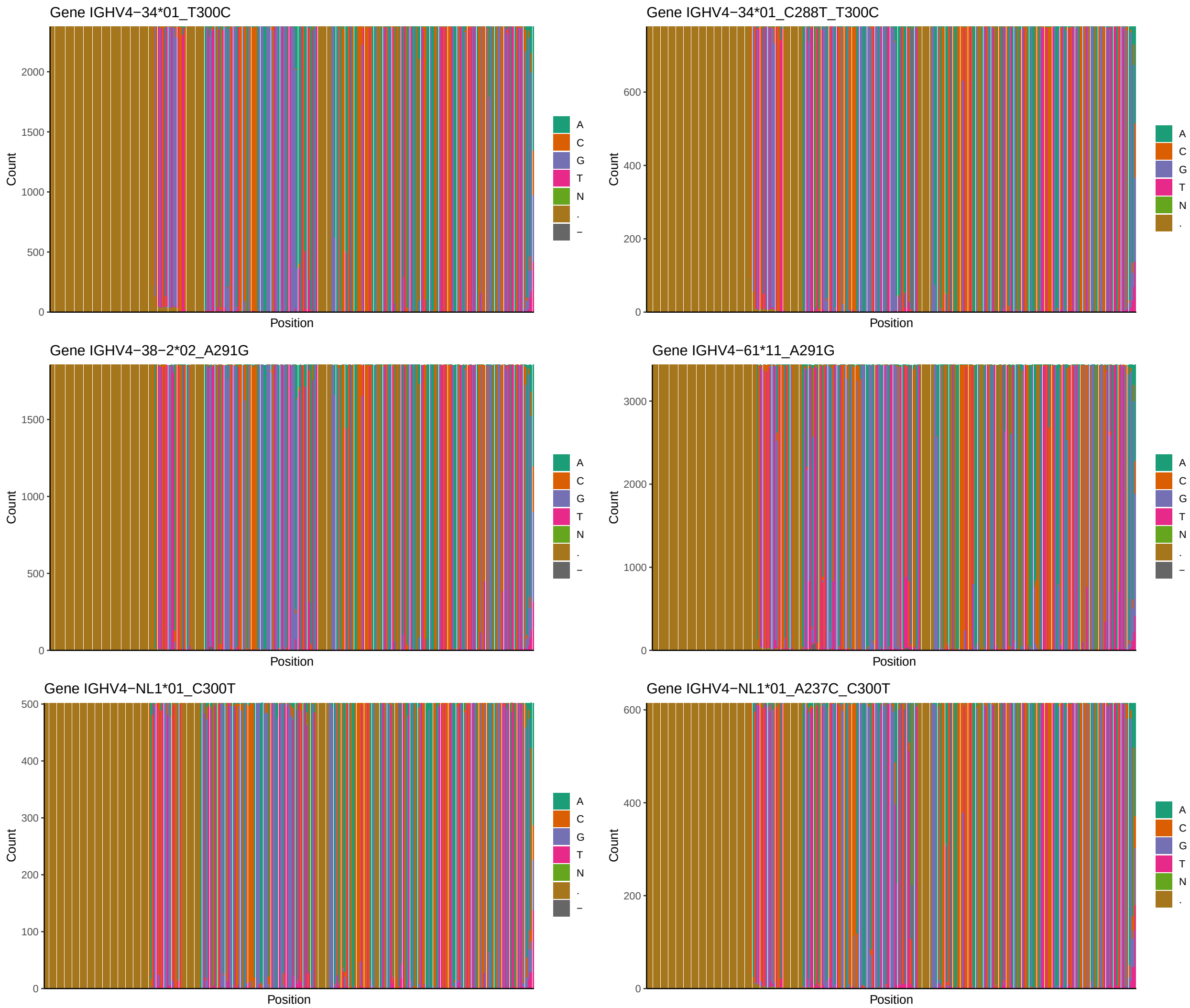
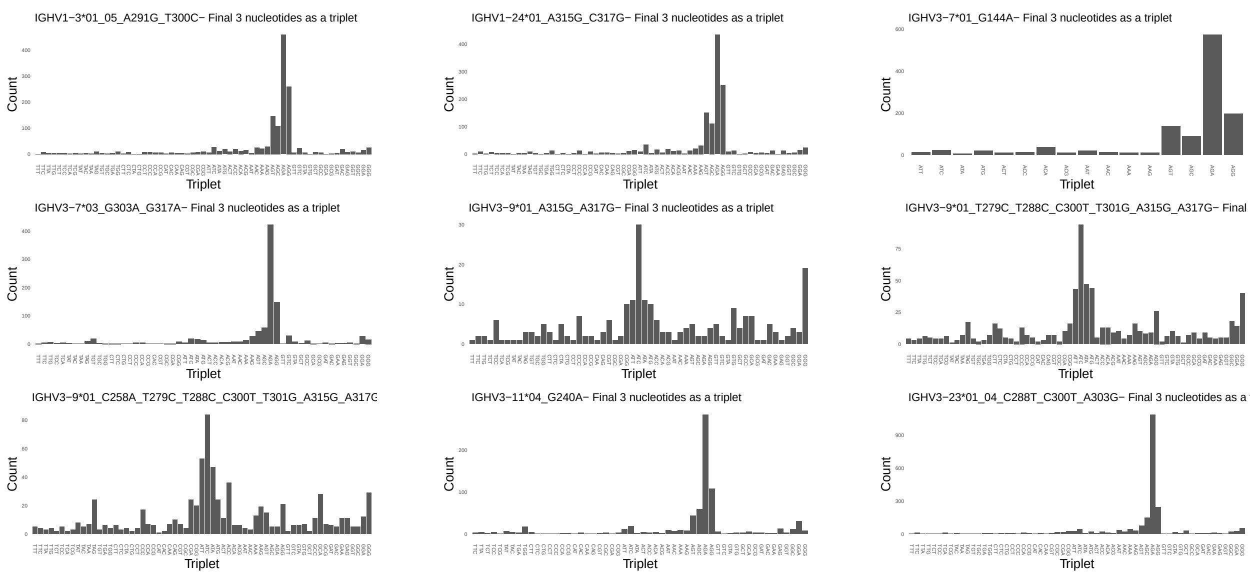
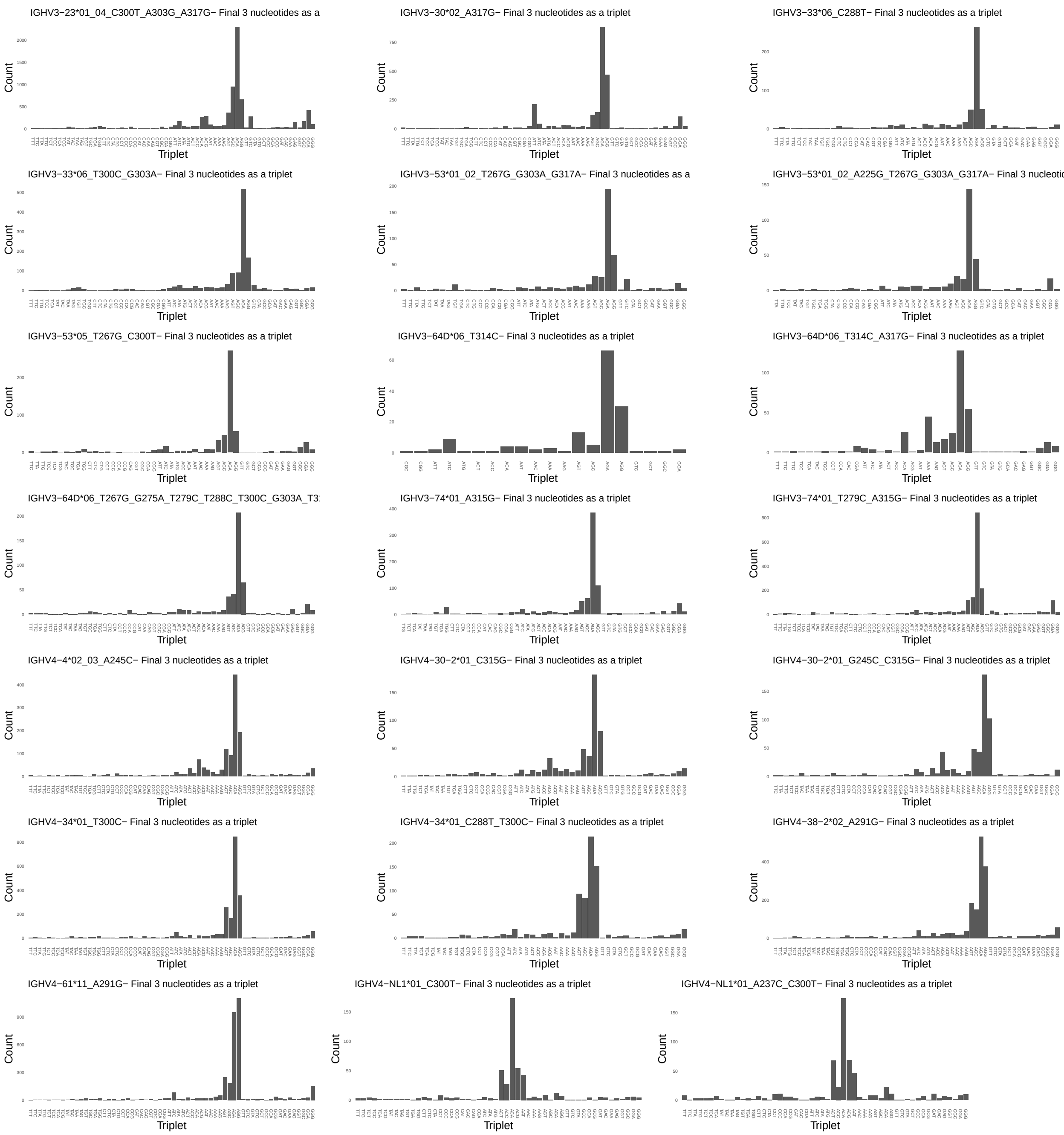

Gene IGHV3−64D*06_T267G_G275A_T279C_T288C_T300C_G303A_T314C
Count
500
400
300
200
100
0
Position
A
C
G
T
N
.

Gene IGHV3−74*01_A315G
Count
750
500
250
0
Position
A
C
G
T
N
.

Gene IGHV3−74*01_T279C_A315G
Count
2000
1500
1000
500
0
Position
A
C
G
T
N
.
−

Gene IGHV4−4*02_03_A245C
Count
1000
500
0
Position
A
C
G
T
N
.
−

Gene IGHV4−30−2*01_C315G
Count
500
400
300
200
100
0
Position
A
C
G
T
N
.
−

Gene IGHV4−30−2*01_G245C_C315G
Count
600
400
200
0
Position
A
C
G
T
N
.
−

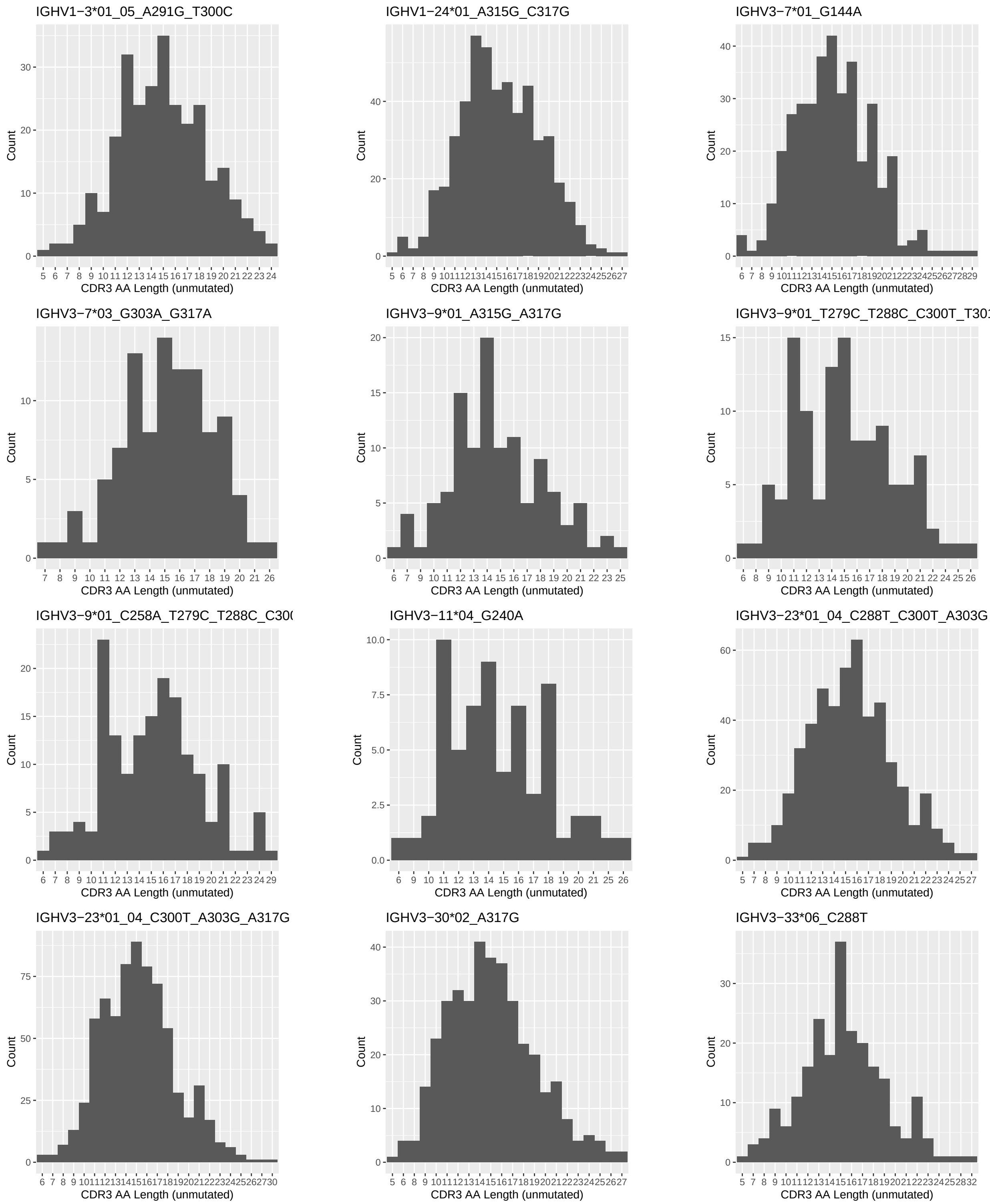


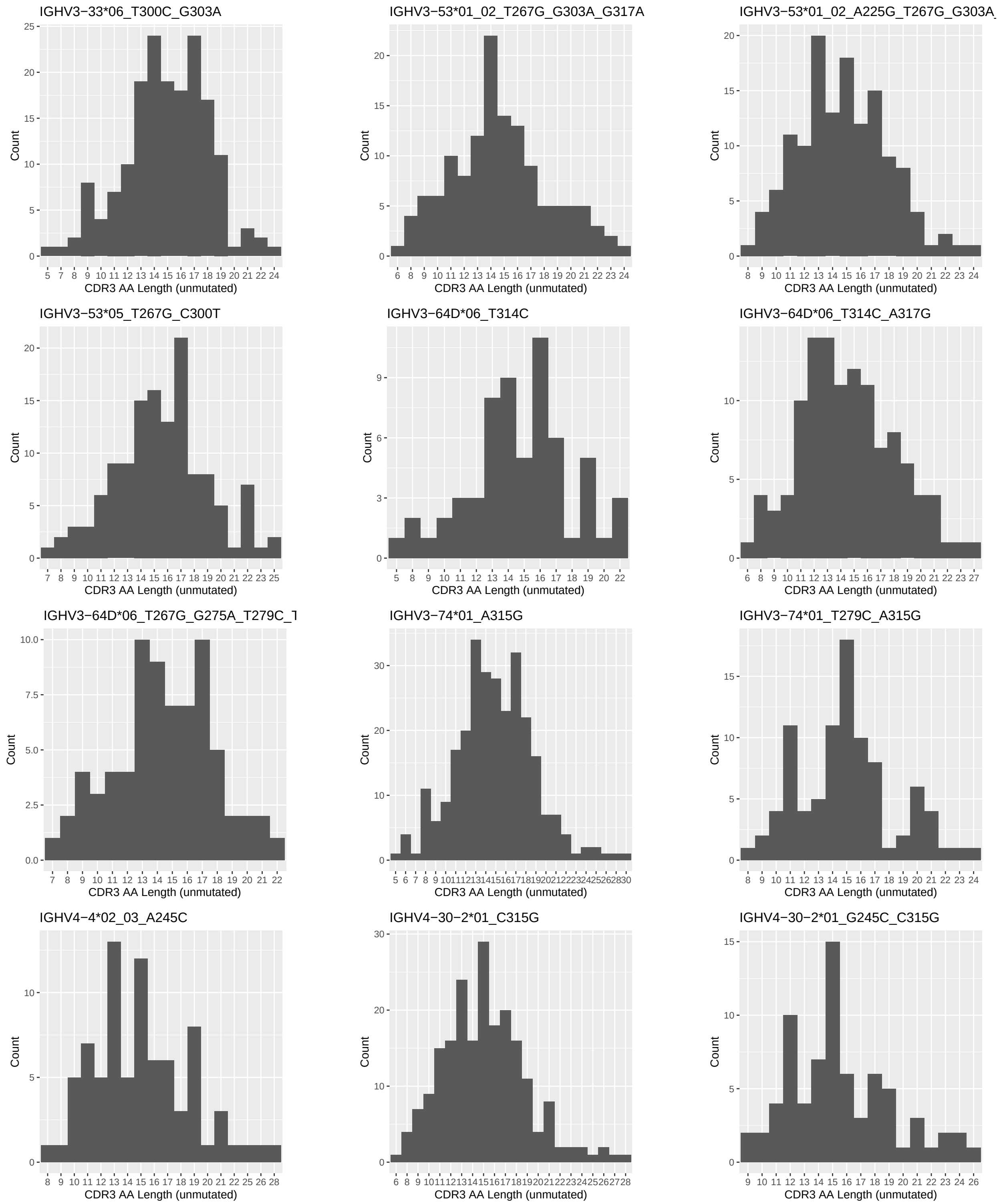
1.4 Final three nucleotides: frequency of each observed triplet

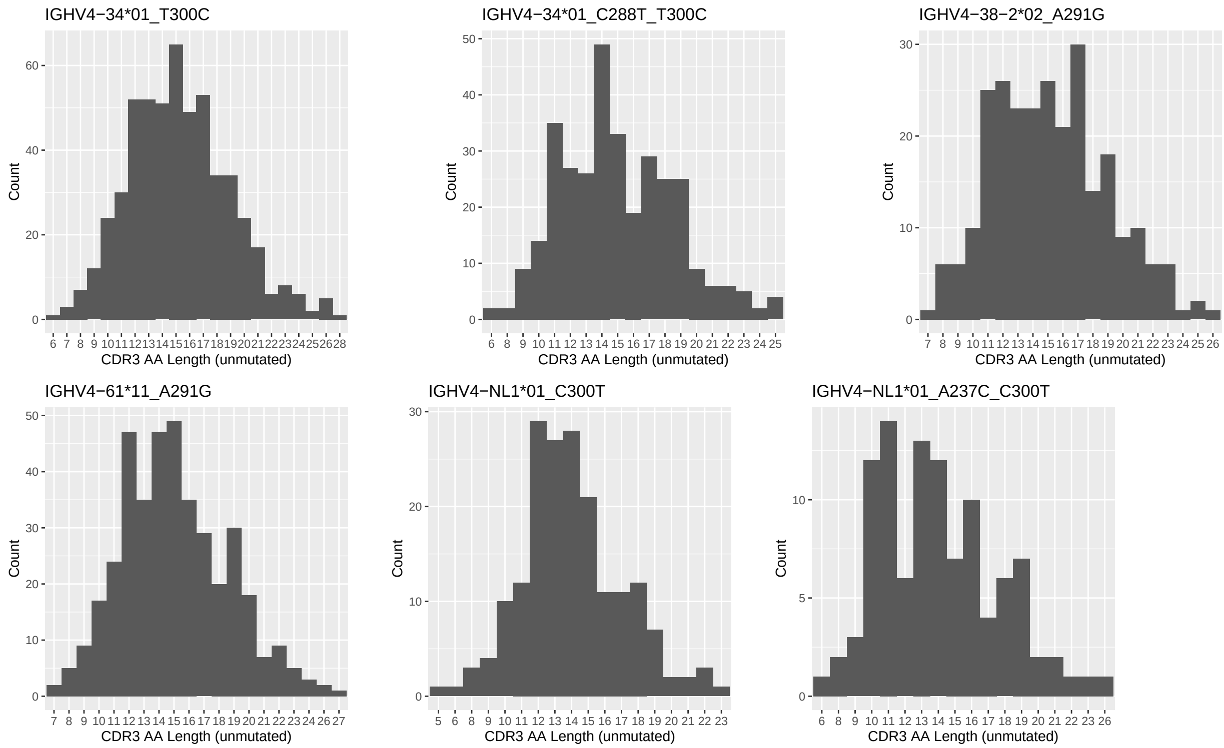



IGHV3−23*01_04_C300T_A303G_A317G− Final 3 nucleotides as a
IGHV3−30*02_A317G− Final 3 nucleotides as a triplet
IGHV3−33*06_C288T− Final 3 nucleotides as a triplet
IGHV3−33*06_T300C_G303A− Final 3 nucleotides as a triplet
IGHV3−53*01_02_T267G_G303A_G317A− Final 3 nucleotides as a
IGHV3−53*01_02_A225G_T267G_G303A_G317A− Final 3 nucleotic
IGHV3−53*05_T267G_C300T− Final 3 nucleotides as a triplet
IGHV3−64D*06_T314C− Final 3 nucleotides as a triplet
IGHV3−64D*06_T314C_A317G− Final 3 nucleotides as a triplet
IGHV3−64D*06_T267G_G275A_T279C_T288C_T300C_G303A_T3
IGHV3−74*01_A315G− Final 3 nucleotides as a triplet
IGHV3−74*01_T279C_A315G− Final 3 nucleotides as a triplet
IGHV4−4*02_03_A245C− Final 3 nucleotides as a triplet
IGHV4−30−2*01_C315G− Final 3 nucleotides as a triplet
IGHV4−30−2*01_G245C_C315G− Final 3 nucleotides as a triplet
IGHV4−34*01_T300C− Final 3 nucleotides as a triplet
IGHV4−34*01_C288T_T300C− Final 3 nucleotides as a triplet
IGHV4−38−2*02_A291G− Final 3 nucleotides as a triplet
IGHV4−61*11_A291G− Final 3 nucleotides as a triplet
IGHV4−NL1*01_C300T− Final 3 nucleotides as a triplet
IGHV4−NL1*01_A237C_C300T− Final 3 nucleotides as a triplet
Count
Triplet

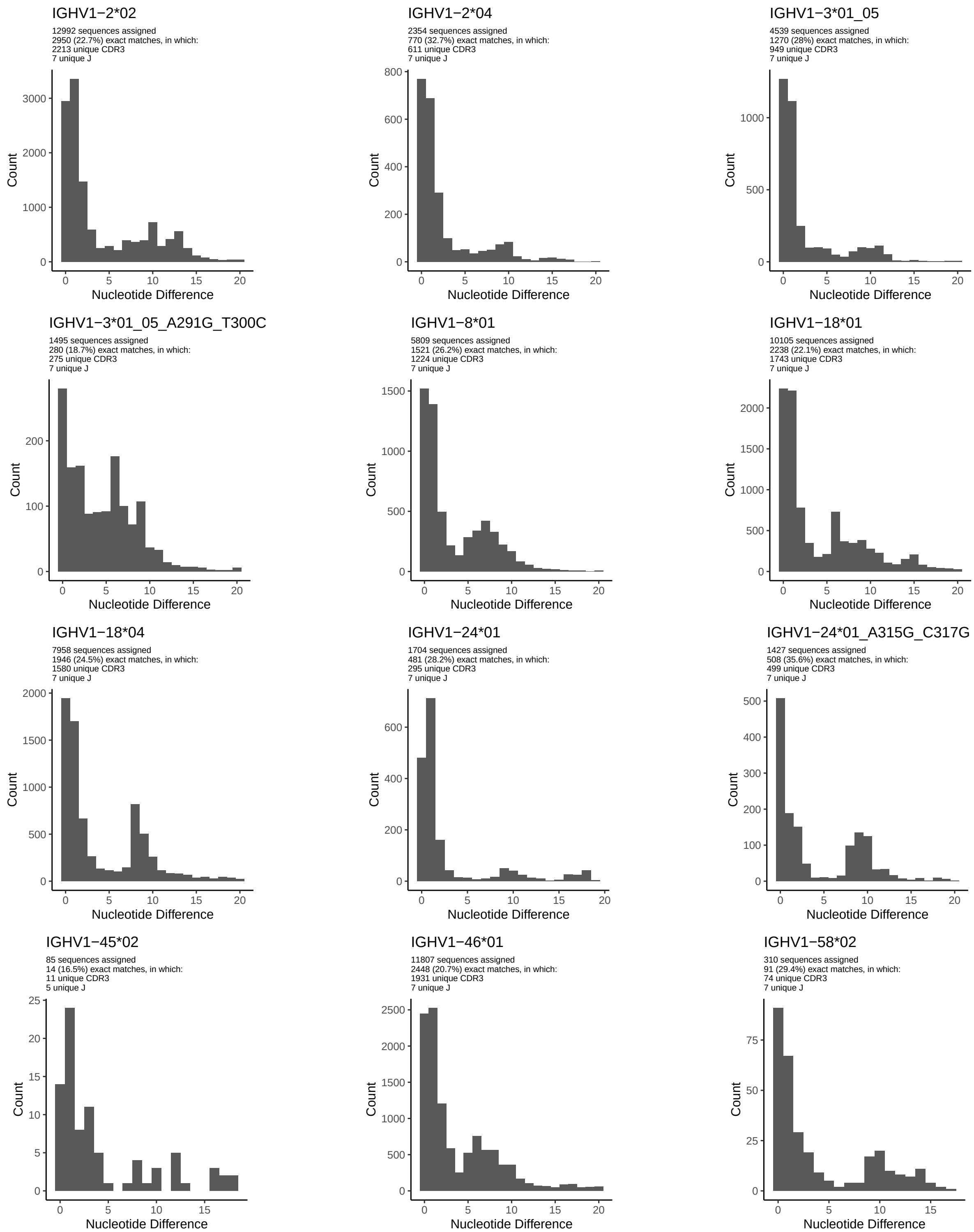
1.5 CDR3 length distribution, in assignments to novel alleles

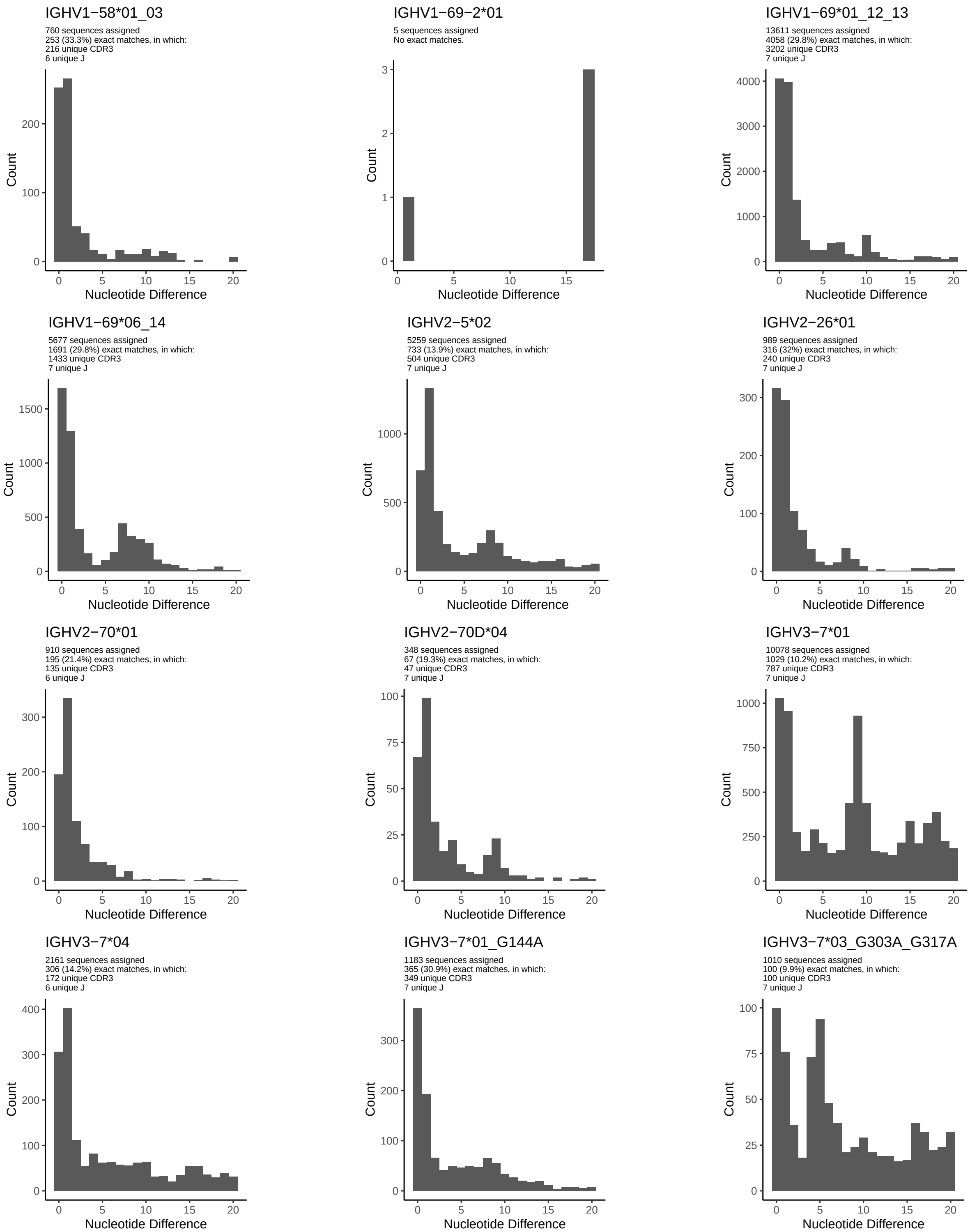


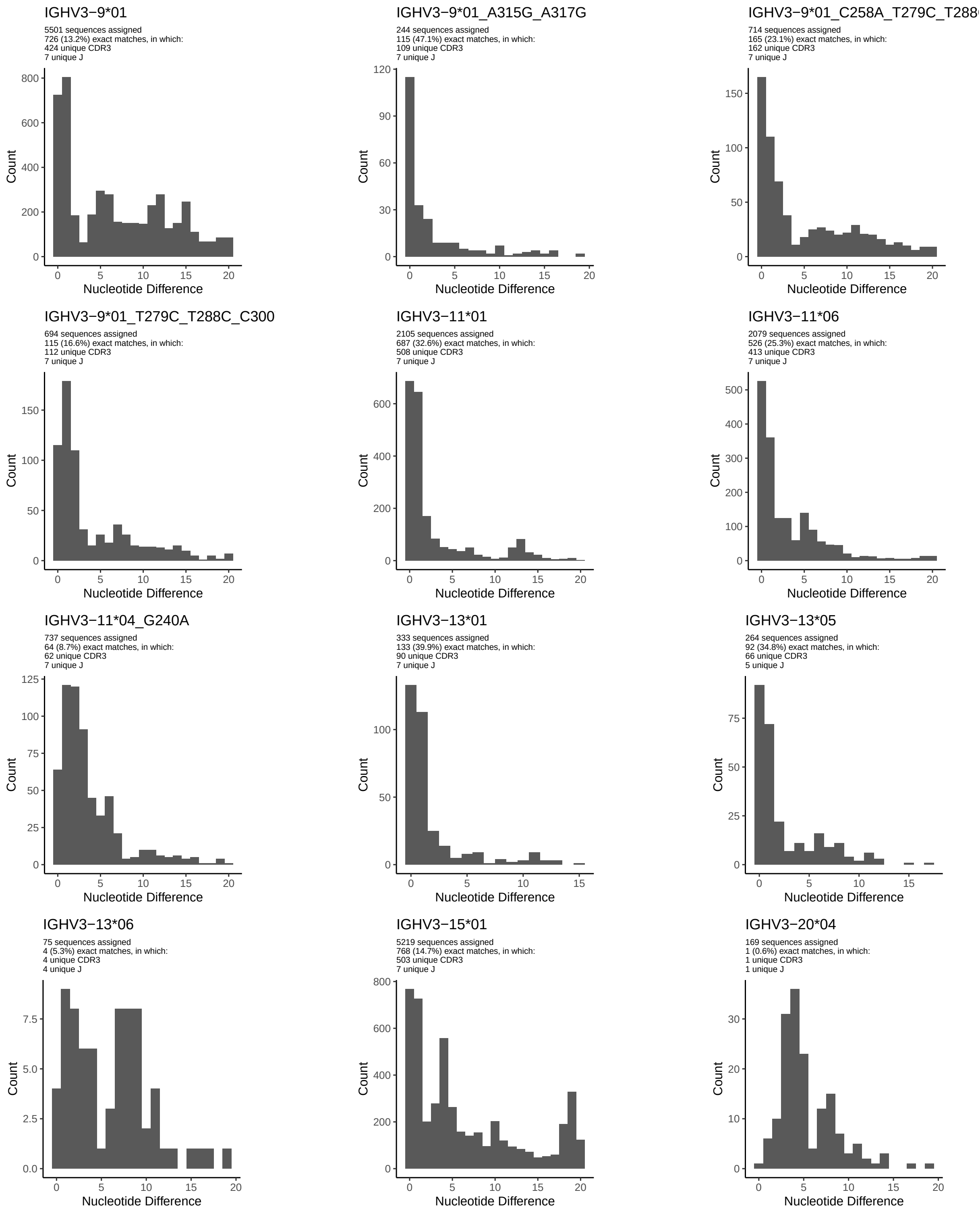

IGHV3−33*06_T300C_G303A
IGHV3−53*01_02_T267G_G303A_G317A
IGHV3−53*01_02_A225G_T267G_G303A_
IGHV3−53*05_T267G_C300T
IGHV3−64D*06_T314C
IGHV3−64D*06_T314C_A317G
IGHV3−64D*06_T267G_G275A_T279C_T
IGHV3−74*01_A315G
IGHV3−74*01_T279C_A315G
IGHV4−4*02_03_A245C
IGHV4−30−2*01_C315G
IGHV4−30−2*01_G245C_C315G
Count
CDR3 AA Length (unmutated)

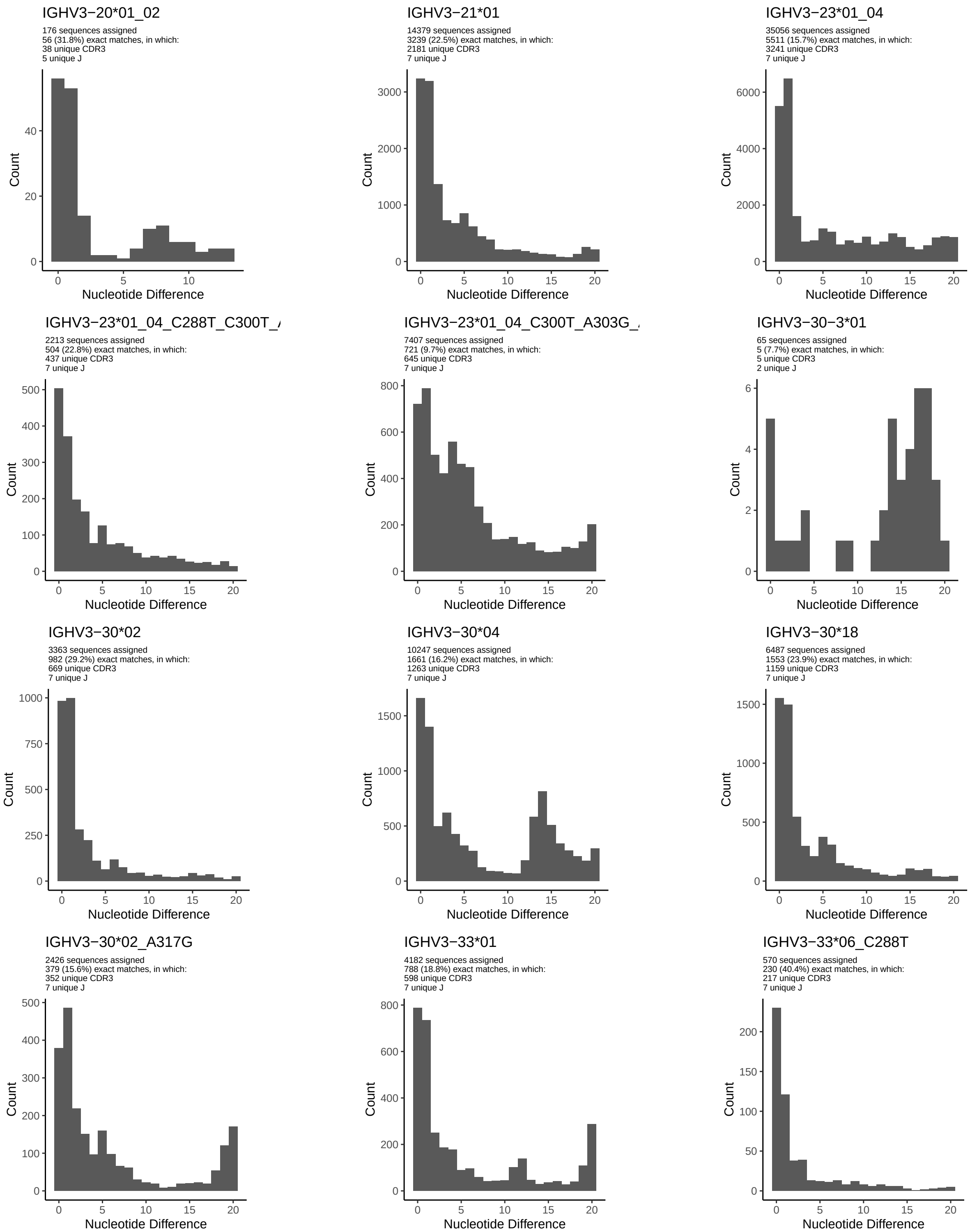

IGHV4−34*01_T300C
IGHV4−34*01_C288T_T300C
IGHV4−38−2*02_A291G
IGHV4−61*11_A291G
IGHV4−NL1*01_C300T
IGHV4−NL1*01_A237C_C300T
Count
CDR3 AA Length (unmutated)

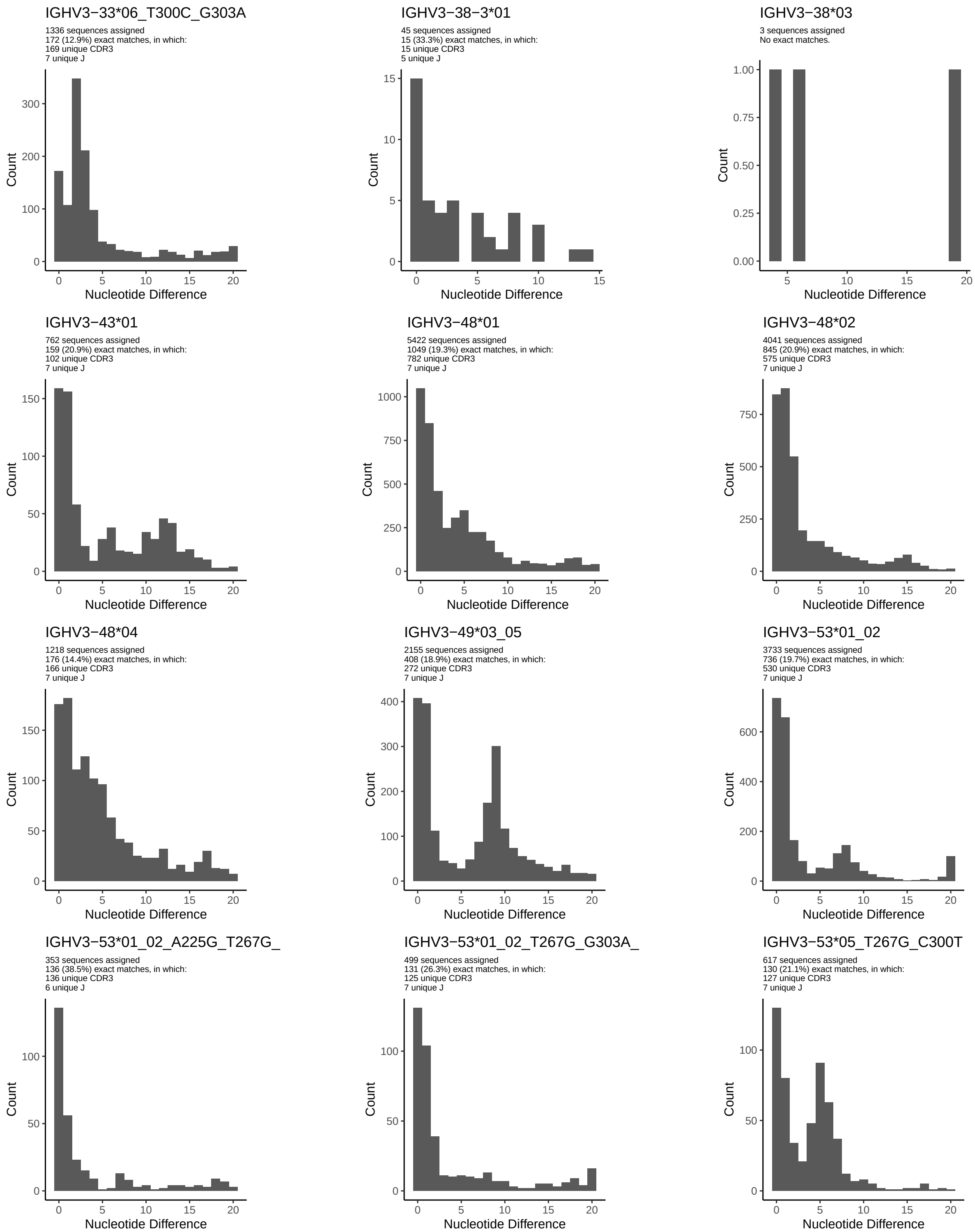
2 Variation from germline, in assignments to each allele

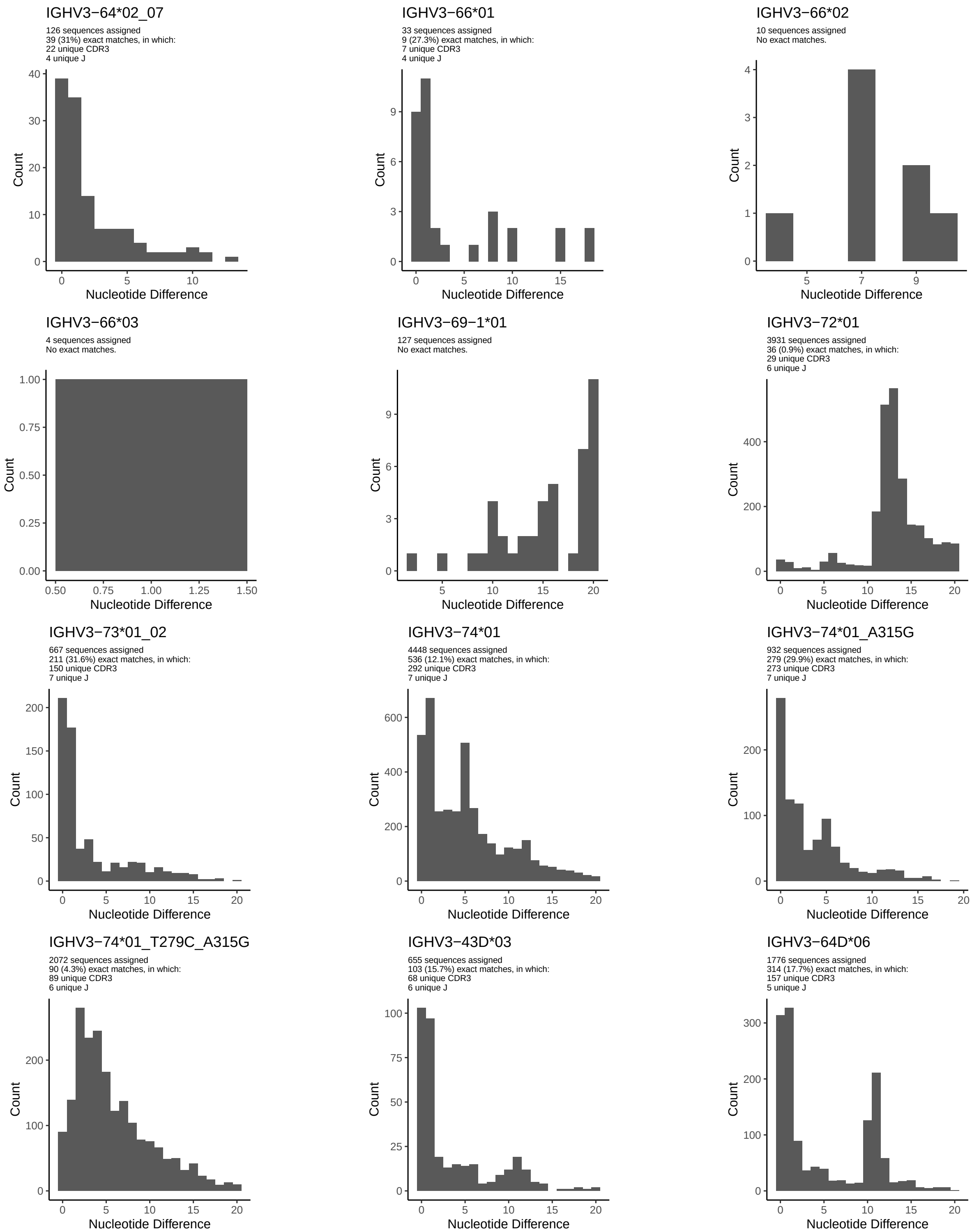


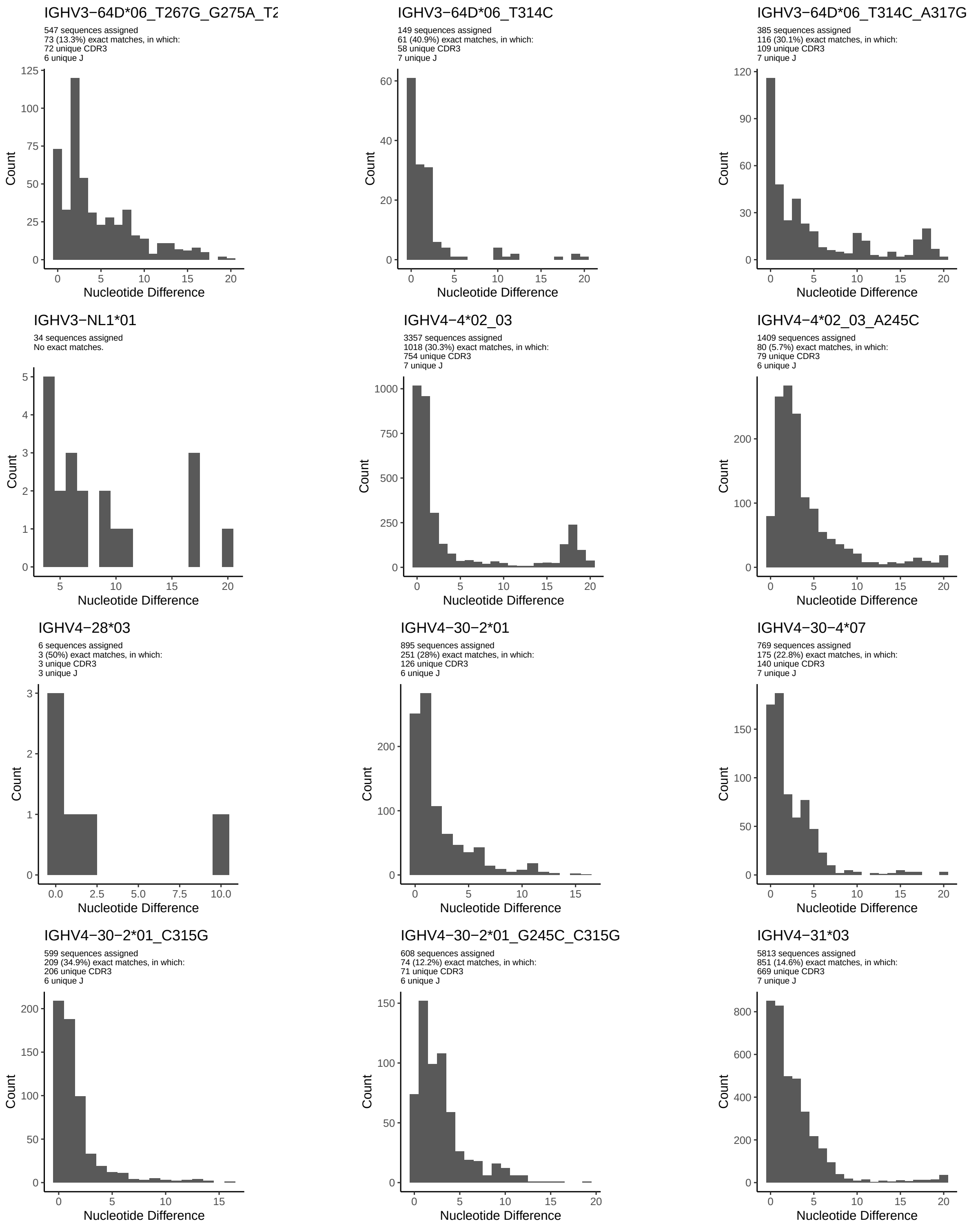

IGHV1−58*01_03
760 sequences assigned
253 (33.3%) exact matches, in which:
216 unique CDR3
6 unique J
IGHV1−69−2*01
5 sequences assigned
No exact matches.
IGHV1−69*01_12_13
13611 sequences assigned
4058 (29.8%) exact matches, in which:
3202 unique CDR3
7 unique J
IGHV1−69*06_14
5677 sequences assigned
1691 (29.8%) exact matches, in which:
1433 unique CDR3
7 unique J
IGHV2−5*02
5259 sequences assigned
733 (13.9%) exact matches, in which:
504 unique CDR3
7 unique J
IGHV2−26*01
989 sequences assigned
316 (32%) exact matches, in which:
240 unique CDR3
7 unique J
IGHV2−70*01
910 sequences assigned
195 (21.4%) exact matches, in which:
135 unique CDR3
6 unique J
IGHV2−70D*04
348 sequences assigned
67 (19.3%) exact matches, in which:
47 unique CDR3
7 unique J
IGHV3−7*01
10078 sequences assigned
1029 (10.2%) exact matches, in which:
787 unique CDR3
7 unique J
IGHV3−7*04
2161 sequences assigned
306 (14.2%) exact matches, in which:
172 unique CDR3
6 unique J
IGHV3−7*01_G144A
1183 sequences assigned
365 (30.9%) exact matches, in which:
349 unique CDR3
7 unique J
IGHV3−7*03_G303A_G317A
1010 sequences assigned
100 (9.9%) exact matches, in which:
100 unique CDR3
7 unique J
Count
Nucleotide Difference

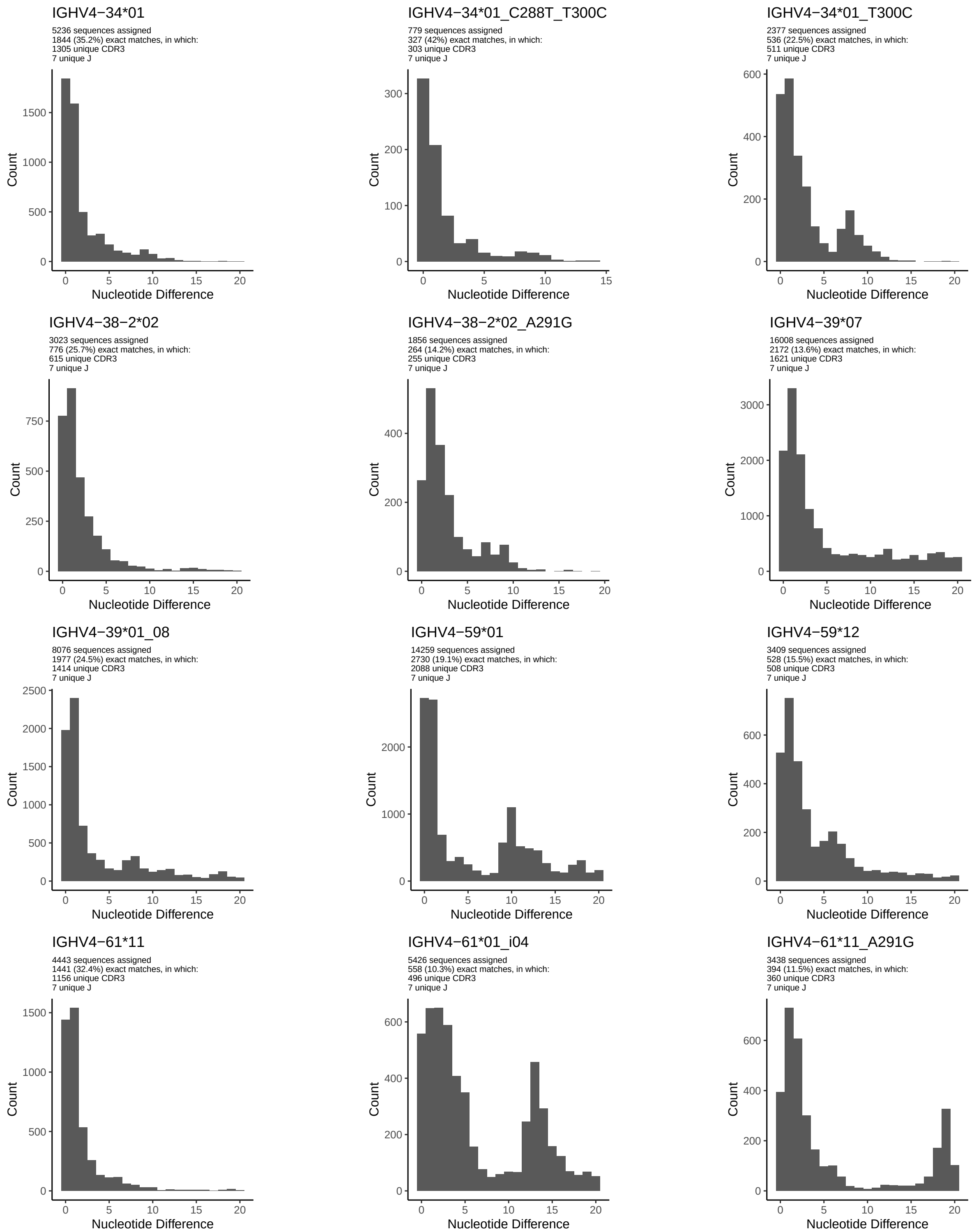

IGHV3−9*01
5501 sequences assigned
726 (13.2%) exact matches, in which:
424 unique CDR3
7 unique J
IGHV3−9*01_A315G_A317G
244 sequences assigned
115 (47.1%) exact matches, in which:
109 unique CDR3
7 unique J
IGHV3−9*01_C258A_T279C_T288
714 sequences assigned
165 (23.1%) exact matches, in which:
162 unique CDR3
7 unique J
IGHV3−9*01_T279C_T288C_C300
694 sequences assigned
115 (16.6%) exact matches, in which:
112 unique CDR3
7 unique J
IGHV3−11*01
2105 sequences assigned
687 (32.6%) exact matches, in which:
508 unique CDR3
7 unique J
IGHV3−11*06
2079 sequences assigned
526 (25.3%) exact matches, in which:
413 unique CDR3
7 unique J
IGHV3−11*04_G240A
737 sequences assigned
64 (8.7%) exact matches, in which:
62 unique CDR3
7 unique J
IGHV3−13*01
333 sequences assigned
133 (39.9%) exact matches, in which:
90 unique CDR3
7 unique J
IGHV3−13*05
264 sequences assigned
92 (34.8%) exact matches, in which:
66 unique CDR3
5 unique J
IGHV3−13*06
75 sequences assigned
4 (5.3%) exact matches, in which:
4 unique CDR3
4 unique J
IGHV3−15*01
5219 sequences assigned
768 (14.7%) exact matches, in which:
503 unique CDR3
7 unique J
IGHV3−20*04
169 sequences assigned
1 (0.6%) exact matches, in which:
1 unique CDR3
1 unique J
Count
Nucleotide Difference

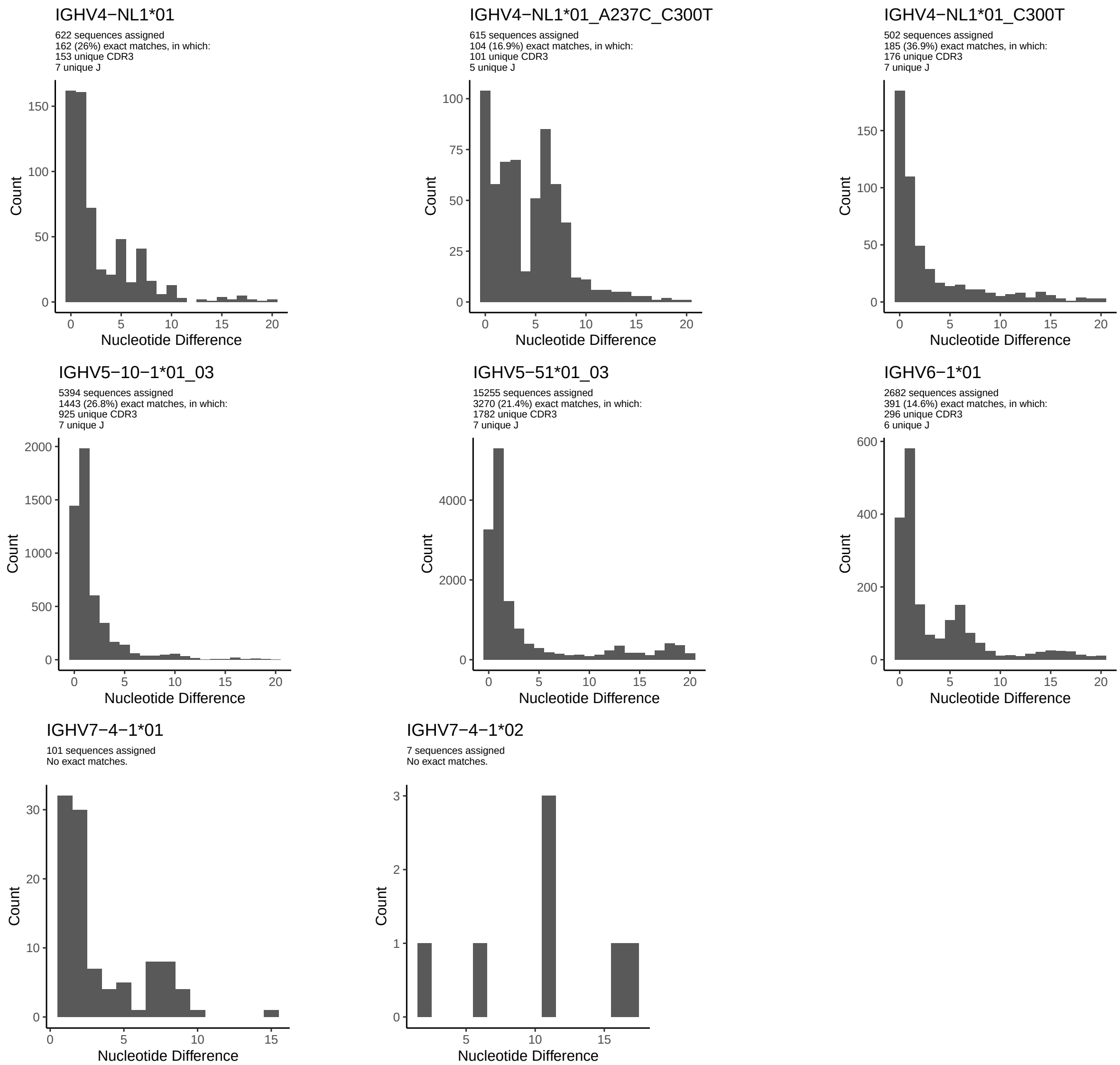

IGHV3−20*01_02
176 sequences assigned
56 (31.8%) exact matches, in which:
38 unique CDR3
5 unique J
IGHV3−21*01
14379 sequences assigned
3239 (22.5%) exact matches, in which:
2181 unique CDR3
7 unique J
IGHV3−23*01_04
35056 sequences assigned
5511 (15.7%) exact matches, in which:
3241 unique CDR3
7 unique J
IGHV3−23*01_04_C288T_C300T_
2213 sequences assigned
504 (22.8%) exact matches, in which:
437 unique CDR3
7 unique J
IGHV3−23*01_04_C300T_A303G_
7407 sequences assigned
721 (9.7%) exact matches, in which:
645 unique CDR3
7 unique J
IGHV3−30−3*01
65 sequences assigned
5 (7.7%) exact matches, in which:
5 unique CDR3
2 unique J
IGHV3−30*02
3363 sequences assigned
982 (29.2%) exact matches, in which:
669 unique CDR3
7 unique J
IGHV3−30*04
10247 sequences assigned
1661 (16.2%) exact matches, in which:
1263 unique CDR3
7 unique J
IGHV3−30*18
6487 sequences assigned
1553 (23.9%) exact matches, in which:
1159 unique CDR3
7 unique J
IGHV3−30*02_A317G
2426 sequences assigned
379 (15.6%) exact matches, in which:
352 unique CDR3
7 unique J
IGHV3−33*01
4182 sequences assigned
788 (18.8%) exact matches, in which:
598 unique CDR3
7 unique J
IGHV3−33*06_C288T
570 sequences assigned
230 (40.4%) exact matches, in which:
217 unique CDR3
7 unique J
Count
Nucleotide Difference


IGHV3−33*06_T300C_G303A
1336 sequences assigned
172 (12.9%) exact matches, in which:
169 unique CDR3
7 unique J
IGHV3−38−3*01
45 sequences assigned
15 (33.3%) exact matches, in which:
15 unique CDR3
5 unique J
IGHV3−38*03
3 sequences assigned
No exact matches.
IGHV3−43*01
762 sequences assigned
159 (20.9%) exact matches, in which:
102 unique CDR3
7 unique J
IGHV3−48*01
5422 sequences assigned
1049 (19.3%) exact matches, in which:
782 unique CDR3
7 unique J
IGHV3−48*02
4041 sequences assigned
845 (20.9%) exact matches, in which:
575 unique CDR3
7 unique J
IGHV3−48*04
1218 sequences assigned
176 (14.4%) exact matches, in which:
166 unique CDR3
7 unique J
IGHV3−49*03_05
2155 sequences assigned
408 (18.9%) exact matches, in which:
272 unique CDR3
7 unique J
IGHV3−53*01_02
3733 sequences assigned
736 (19.7%) exact matches, in which:
530 unique CDR3
7 unique J
IGHV3−53*01_02_A225G_T267G_
353 sequences assigned
136 (38.5%) exact matches, in which:
136 unique CDR3
6 unique J
IGHV3−53*01_02_T267G_G303A_
499 sequences assigned
131 (26.3%) exact matches, in which:
125 unique CDR3
7 unique J
IGHV3−53*05_T267G_C300T
617 sequences assigned
130 (21.1%) exact matches, in which:
127 unique CDR3
7 unique J
Count
Nucleotide Difference

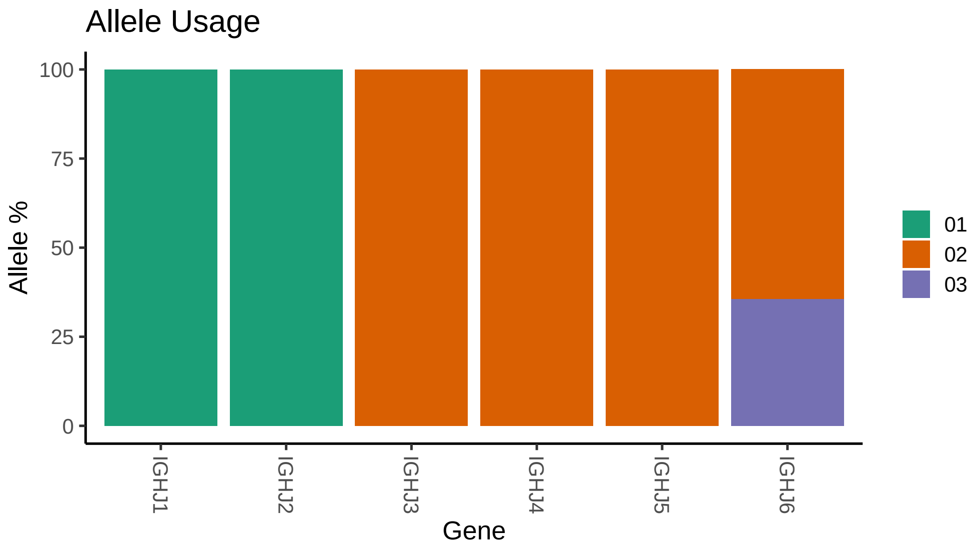

IGHV3−64*02_07
126 sequences assigned
39 (31%) exact matches, in which:
22 unique CDR3
4 unique J
IGHV3−66*01
33 sequences assigned
9 (27.3%) exact matches, in which:
7 unique CDR3
4 unique J
IGHV3−66*02
10 sequences assigned
No exact matches.
IGHV3−66*03
4 sequences assigned
No exact matches.
IGHV3−69−1*01
127 sequences assigned
No exact matches.
IGHV3−72*01
3931 sequences assigned
36 (0.9%) exact matches, in which:
29 unique CDR3
6 unique J
IGHV3−73*01_02
667 sequences assigned
211 (31.6%) exact matches, in which:
150 unique CDR3
7 unique J
IGHV3−74*01
4448 sequences assigned
536 (12.1%) exact matches, in which:
292 unique CDR3
7 unique J
IGHV3−74*01_A315G
932 sequences assigned
279 (29.9%) exact matches, in which:
273 unique CDR3
7 unique J
IGHV3−74*01_T279C_A315G
2072 sequences assigned
90 (4.3%) exact matches, in which:
89 unique CDR3
6 unique J
IGHV3−43D*03
655 sequences assigned
103 (15.7%) exact matches, in which:
68 unique CDR3
6 unique J
IGHV3−64D*06
1776 sequences assigned
314 (17.7%) exact matches, in which:
157 unique CDR3
5 unique J
Count
Nucleotide Difference


IGHV3−64D*06_T267G_G275A_T2
547 sequences assigned
73 (13.3%) exact matches, in which:
72 unique CDR3
6 unique J
IGHV3−64D*06_T314C
149 sequences assigned
61 (40.9%) exact matches, in which:
58 unique CDR3
7 unique J
IGHV3−64D*06_T314C_A317G
385 sequences assigned
116 (30.1%) exact matches, in which:
109 unique CDR3
7 unique J
IGHV3−NL1*01
34 sequences assigned
No exact matches.
IGHV4−4*02_03
3357 sequences assigned
1018 (30.3%) exact matches, in which:
754 unique CDR3
7 unique J
IGHV4−4*02_03_A245C
1409 sequences assigned
80 (5.7%) exact matches, in which:
79 unique CDR3
6 unique J
IGHV4−28*03
6 sequences assigned
3 (50%) exact matches, in which:
3 unique CDR3
3 unique J
IGHV4−30−2*01
895 sequences assigned
251 (28%) exact matches, in which:
126 unique CDR3
6 unique J
IGHV4−30−4*07
769 sequences assigned
175 (22.8%) exact matches, in which:
140 unique CDR3
7 unique J
IGHV4−30−2*01_C315G
599 sequences assigned
209 (34.9%) exact matches, in which:
206 unique CDR3
6 unique J
IGHV4−30−2*01_G245C_C315G
608 sequences assigned
74 (12.2%) exact matches, in which:
71 unique CDR3
6 unique J
IGHV4−31*03
5813 sequences assigned
851 (14.6%) exact matches, in which:
669 unique CDR3
7 unique J
Count
Nucleotide Difference


IGHV4−34*01
5236 sequences assigned
1844 (35.2%) exact matches, in which:
1305 unique CDR3
7 unique J
IGHV4−34*01_C288T_T300C
779 sequences assigned
327 (42%) exact matches, in which:
303 unique CDR3
7 unique J
IGHV4−34*01_T300C
2377 sequences assigned
536 (22.5%) exact matches, in which:
511 unique CDR3
7 unique J
IGHV4−38−2*02
3023 sequences assigned
776 (25.7%) exact matches, in which:
615 unique CDR3
7 unique J
IGHV4−38−2*02_A291G
1856 sequences assigned
264 (14.2%) exact matches, in which:
255 unique CDR3
7 unique J
IGHV4−39*07
16008 sequences assigned
2172 (13.6%) exact matches, in which:
1621 unique CDR3
7 unique J
IGHV4−39*01_08
8076 sequences assigned
1977 (24.5%) exact matches, in which:
1414 unique CDR3
7 unique J
IGHV4−59*01
14259 sequences assigned
2730 (19.1%) exact matches, in which:
2088 unique CDR3
7 unique J
IGHV4−59*12
3409 sequences assigned
528 (15.5%) exact matches, in which:
508 unique CDR3
7 unique J
IGHV4−61*11
4443 sequences assigned
1441 (32.4%) exact matches, in which:
1156 unique CDR3
7 unique J
IGHV4−61*01_i04
5426 sequences assigned
558 (10.3%) exact matches, in which:
496 unique CDR3
7 unique J
IGHV4−61*11_A291G
3438 sequences assigned
394 (11.5%) exact matches, in which:
360 unique CDR3
7 unique J
Count
Nucleotide Difference


IGHV4−NL1*01
622 sequences assigned
162 (26%) exact matches, in which:
153 unique CDR3
7 unique J
IGHV4−NL1*01_A237C_C300T
615 sequences assigned
104 (16.9%) exact matches, in which:
101 unique CDR3
5 unique J
IGHV4−NL1*01_C300T
502 sequences assigned
185 (36.9%) exact matches, in which:
176 unique CDR3
7 unique J
IGHV5−10−1*01_03
5394 sequences assigned
1443 (26.8%) exact matches, in which:
925 unique CDR3
7 unique J
IGHV5−51*01_03
15255 sequences assigned
3270 (21.4%) exact matches, in which:
1782 unique CDR3
7 unique J
IGHV6−1*01
2682 sequences assigned
391 (14.6%) exact matches, in which:
296 unique CDR3
6 unique J
IGHV7−4−1*01
101 sequences assigned
No exact matches.
IGHV7−4−1*02
7 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /work/jenkins/PRJNA491287/M64 010/M64 010/M64 010/M64 010/9d/c1a418b358e760f163eab2
##
## Germline reference file: /work/jenkins/PRJNA491287/M64 010/M64 010/M64 010/M64 010/9d/c1a418b358e760
##
## Novel allele file: /work/jenkins/PRJNA491287/M64 010/M64 010/M64 010/M64 010/9d/c1a418b358e760f163ea
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 7 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 7 01_G144A: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 7 03_G303A_G317A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 9 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 9 01_A315G_A317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 9 01_T279C_T288C_C300T_T301G_A315G_A317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 9 01_C258A_T279C_T288C_C300T_T301G_A315G_A317G: replacing with
##
##
## Unexpected N in reference sequence IGHV3 11 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 11 04_G240A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C288T_C300T_A303G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 02_A317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_C288T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 53 01_02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 53 01_02_T267G_G303A_G317A: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 53 01_02_A225G_T267G_G303A_G317A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 66 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 53 05_T267G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 64D 06: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 64D 06_T314C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 64D 06_T314C_A317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 64D 06_T267G_G275A_T279C_T288C_T300C_G303A_T314C: replacing wit
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 74 01_T279C_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_G245C_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_C288T_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 38 2 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 02_A291G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 61 11: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 11_A291G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_A237C_C300T: replacing with gap
```